Math 6230 – Partial Differential Equations II– Spring 2026
w/Professor Perrera

January 12, 2026: Hölder semi-norm

Hölder Spaces

let $U \subset \mathbb{R}^n$ be an open se.t

Definition: let $C(\overline{U})$ denote the linear space of all continuous and bounded functions $u : U \to \mathbb{R}$. For $u \in C(\overline{U})$, define

$$||u||_{C(\overline{U})} = \sup_{x \in U} |u(x)|$$

Theorem: $|| \bullet ||_{C(\overline{U})}$ is a norm on $C(\overline{U})$.

Proof:

1. Clearly, $||u||_{C(\overline{U})} \geq 0$ and equaltiy holds iff $u(x) = 0$ for all $x \in U$, i.e., u=0.
2. For $u \in C(\overline{U})$ and $\lambda \in \mathbb{R}$

$$||\lambda u||_{C(\overline{U})} = |\lambda| \, ||u||_{C(\overline{U})} = |\lambda| \left(\sup_{x \in U} |u(x)| \right) |\lambda| \, ||u||_{C(\overline{U})}$$

3. For $u, v \in C(\overline{U})$ and $x \in U$

$$\begin{aligned} |u(x) + v(x)| &\leq |u(x)| + |v(x)| \\ &\leq \sup_{y \in u} |u(y)| + \sup_{y \in u} |v(y)| \\ &= \underbrace{||u||_{C(\overline{U})} + ||v||_{C(\overline{U})}}_{\text{constant}} \end{aligned}$$

Taking the sup over all $x \in U$ gives

$$\begin{aligned} \sup_{x \in U} |u(x) + v(x)| &\leq ||u||_{C(\overline{U})} + ||u||_{C(\overline{U})} \\ \therefore ||u + v||_{C(\overline{U})} &\leq ||u||_{C(\overline{U})} + ||u||_{C(\overline{U})} \end{aligned}$$

Theorem: $C(\overline{U})$ is complete, i.e., every Cauchy seq $(u_j) \in C(\overline{U})$ converges to some $u \in C(\overline{U})$.

Recall: $u : U \to \mathbb{R}$ is Lipschitz continuous if $|u(x) - u(y)| \leq C|x - y|, \forall x, y \in U$ for some constant $C > 0$.

Definition: We say that $u : U \to \mathbb{R}$ is Hölder continuous, with Hölder exponent $\gamma \in (0, 1]$ if

$$|u(x) - u(y)| \leq C|x - y|^\gamma, \, \forall x, y \in U$$

for some constant $C > 0$ ($\gamma = 1$ corresponds to Lipschitz continuity). Then

$$[u]_{C^{0,\gamma}(\overline{U})} = \sup_{x, y \in U, x \neq y} \frac{|u(x) - u(y)|}{|x - y|^\gamma}$$

as the Hölder seminorm of u.

Theorem

1. $[\lambda u]_{C^{0,\gamma}(\overline{U})} = |\lambda|\,[u]_{C^{0,\gamma}(\overline{U})}$,
2. $[u+v]_{C^{0,\gamma}(\overline{U})} \leq [u]_{C^{0,\gamma}(\overline{U})} + [v]_{C^{0,\gamma}(\overline{U})}$

Proof is mostly plug and play.

Definition: Let $C^{0,\gamma}(\overline{U})$ denote the subspace of $C(\overline{U})$ consisting of functions u for which $[u]_{C^{0,\gamma}(\overline{u})} < \infty$. For $u \in C^{0,\gamma}(\overline{U})$ define the ***Hölder Norm*** of u by

$$||u||_{C^{0,\gamma}(\overline{U})} = ||u||_{C(\overline{U})} + [u]_{C^{0,\gamma}(\overline{U})}$$

Theorem: $||\bullet||_{C^{0,\gamma}(\overline{U})}$ is a norm on $C^{0,\gamma}(\overline{U})$.

Theorem: $C^{0,\gamma}(\overline{U})$ is complete, i.e., it is a Banach space.

Janary 14, 2026: Proof of Hölder Space $\in$ Banach Space

The proof goes from norms on $C(\overline{U})$ to $C^1(\overline{U})$ and upward. Transcribe from online notes.

1. $||\bullet||_{C^{0,\gamma}(\overline{U})}$

$$||u||_{C^{0,\gamma}(\overline{U})} = ||u||_{C(\overline{U})} + [u]_{C^{0,\gamma}(\overline{U})}$$

2. $||\bullet||_{C^1(\overline{U})}$

   Definition: Let $C^1(\overline{U})$ denote the linear space of continuously differential functions $u : U \to \mathbb{R}$ such that

$$||u||_{C^1(\overline{U})} := ||u||_{C(\overline{U})} + \sum_{i=1}^{n} ||u_{x_i}||_{C(\overline{U})} < \infty$$

   where $u_{x_i} = \frac{\partial u}{\partial x_i}$.

   Theorem: $||\bullet||_{C^1(\overline{U})}$ is a norm on $C^1(\overline{U})$.

   Theorem: $C^1(\overline{U})$ is a Banach space.

3. $||\bullet||_{C^{1,\gamma}(U)}$

   Definition: Let $C^{1,\gamma}(\overline{U})$ denote the space of continuously differentialbe functions $u : U \to \mathbb{R}$ such that

$$||u||_{C^{1,\gamma}(U)} = ||u||_{C^1(\overline{U})} + \sum_{i=1}^{n} +[u_{x_i}]_{C^{0,\gamma}(\overline{U})} < \infty$$

   Theorem: $||\bullet||_{C^{1,\gamma}(U)}$ is a norm on $C^{1,\gamma}(\overline{U})$.

   Thoerem: $C^{1,\gamma}(\overline{U})$ is a Banach space.

4. $||\bullet||_{C^2(\overline{U})}$

$$\begin{aligned} ||u||_{C^2(\overline{U})} &= ||u||_{C(\overline{U})} + \sum_{i=1}^{n} ||u_{x_i}||_{C(\overline{U})} + \sum_{i,j=1}^{n} ||u_{x_i x_j}||_{C(\overline{U})} < \infty \\ &\stackrel{?}{=} ||u||_{C^1(\overline{U})} + \sum_{i,j=1}^{n} ||u_{x_i x_j}||_{C(\overline{U})} \end{aligned}$$

5. $||\bullet||_{C^{2,\gamma}(U)}$

$$||u||_{C^{2,\gamma}(\overline{U})} = ||u||_{C(\overline{U})} + \sum_{i=1}^{n} ||u_{x_i}||_{C(\overline{U})} + \sum_{i,j=1}^{n} ||u_{x_ix_j}||_{C(\overline{U})} + \sum_{i,j=1}^{n} [u_{x_ix_j}]_{C^{0,\gamma}(\overline{U})} < \infty$$
$$\stackrel{?}{=} ||u||_{C^2(\overline{U})} + + \sum_{i,j=1}^{n} [u_{x_ix_j}]_{C^{0,\gamma}(\overline{U})}$$

6. Note: the general case is a list of sums over the permuations of the partials of x_i

$$||u||_{C^{k,\gamma}(\overline{U})} := \sum_{|\alpha|\leq k} ||D^\alpha u||_{C(\overline{U})} + \sum_{|\alpha|=k} [D^\alpha u]_{C^{0,\gamma}(\overline{U})}$$
$$\stackrel{?}{=} ||u||_{C^k(\overline{U})} + \sum_{|\alpha|=k} [D^\alpha u]_{C^{0,\gamma}(\overline{U})}$$

keep in mind *uniform convergence*.

Homework #1

Assume $0 < \beta < \gamma < 1$. Prove the interpolation inequality

$$||u||_{C^{0,\gamma}(U)} \leq ||u||_{C^{0,\beta}(U)}^{\frac{1-\gamma}{1-\beta}} \, ||u||_{C^{0,1}(U)}^{\frac{\gamma-\beta}{1-\beta}}$$

Q: Do the exponents make up Hölder Conjugate terms?

$$\frac{1-\gamma}{1-\beta} + \frac{\gamma-\beta}{1-\beta} = \frac{1-\gamma+\gamma-\beta}{1-\beta} = 1$$

Yes, so let's find

$$\theta = \frac{1-\gamma}{1-\beta} \text{ and } 1-\theta = \frac{\gamma-\beta}{1-\beta}$$

to find Hölder conjugate values for γ and β and we find that

$$\gamma = \theta\beta + (1-\theta)$$

Then, substituting into the original equation we get

$$||u||_{C^{0,\gamma}(U)} \leq ||u||_{C^{0,\beta}(U)}^{\theta} \, ||u||_{C^{0,1}(U)}^{1-\theta}$$

Thoughts: Let's put this in context with Lebesque spaces $L^p(U)$ and $L^q(U)$. Lebesque are described as functions that are finite with integration of the function to the $1/p^{\text{th}}$ power. Further, if $p < q$ then $L^p(U) \subseteq L^q(U)$. Hölder's Inequality deals with the relationship between Hölder conjugate Lebesque spaces.

The Hölder Space is defined as k-differentiable and with finite semi-norms on the kth partial derivatives.

Spoiler: Appendex D (or was it Wikipaedia?) indicates that $L^p(U)$ and $L^q(U)$ are dual spaces of each other. Considering these spaces in terms of functions-as-vectors may be the important distinction.

See "AI Questions below: how are Hölder spaces related to Lebesque Spaces?"

January 21, 2026

Multi-index notation:

$\alpha = (\alpha_1, \ldots, \alpha_n)$ a vector of integers $\alpha_i \in \mathbb{N}$. Then given $u = u(x_1, \ldots, x_n)$. Then we define

$$D^\alpha u = \frac{d^{|\alpha|}u}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}$$

Define the general Hölder Space $C^{k,\gamma}(\overline{U})$ as the space of $u : U \to \mathbb{R}$ that are k-times continuously differentialbe with all k-th order partial derivatives Hölder continuous of exponent γ with the norm

$$||u||_{C^{k,\gamma}(\overline{U})} = \sum_{|\alpha| \leq k} D^\alpha$$

Theorem: $C^{k,\gamma}(\overline{U})$ is a Banach space.

$$C^{2,\gamma}(\overline{U}) = ||u||_{C(\overline{U})} + \sum ||u_x||_{C(\overline{U})} + \sum ||u_{xx}||_{C(\overline{U})} + \sum [u_{xx}]_{C^{2,\gamma}(\overline{U})}$$

Note: the seminorms are not included (see $C^{2,\gamma}(\overline{U})$ expansion) in the list of terms (all of them sums over norms). The seminorms are already included between through the process of intermediate partial derivatives.

Sobolev Spaces

Recall $L^p(\mathbb{R}^n)$ = set of measurable functions $u : \mathbb{R}^n \to \mathbb{R}$ such that $\int_{\mathbb{R}^n} |u|^p \, dx < \infty$. And $L^p_{\text{loc}}(\mathbb{R}^n)$ is the set of measurable functions $u : \mathbb{R}^n \to \mathbb{R}$ such that $\int_V |u|^p \, dx < \infty$ for every open set $V \subset \mathbb{R}^n$.

AND, $L^p_{\text{loc}}(\mathbb{R}^n)$ = set of measurable functions $u : \mathbb{R}^n \to \mathbb{R}$ such that $\int_V |u|^p \, dx < \infty$ for every open set $V \subset \mathbb{R}^n$.

Theorem 0.0.1. *$C^{k,\gamma}(\overline{U})$ is a Banach space.*

Definition 0.0.2. Denote by $C_c^\infty(\mathbb{R}^n)$ the space of smooth functions with acompact supports in $\mathbb{R}^n$. Functions in $C_c^\infty(\mathbb{R}^n)$ are called ***test functions***. (Note: infinitely smooth with a very well defined domain).

Useful identity: Let $u \in C(\overline{U})\mathbb{R}^n$ and let $\varphi \in C_c^\infty(\mathbb{R}^n)$ (a test function). Then

$$\int_{\mathbb{R}^n} u_{x_i} \varphi \, dx = -\int_{\mathbb{R}^n} u \varphi_{x_i} dx.$$

It is easy to see that the usual integral over the boundary will be zero, leaving this result. Iterating this we get

$$\int_{\mathbb{R}^n} \varphi D^\alpha u \, dx = (-1)^{|\alpha|} \int_{\mathbb{R}^n} u D^\alpha \varphi \, dx.$$

Definition 0.0.3. Let $u = L^1_{\text{loc}}(\mathbb{R}^n)$. We say that a function $v \in L^1_{\text{loc}}(\mathbb{R}^n)$ is the αth ***weak partial derivative*** of u if

$$\int_{\mathbb{R}^n} u D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_{\mathbb{R}^n} v \varphi \, dx, \, \forall \varphi \in C_0^\infty(\mathbb{R}^n).$$

Definition 0.0.4. We say that $v = D^\alpha u$ ***weakly*** if

$$\int_U u D^\alpha \varphi \, dx = (-1)^{|\alpha|} \int_U v \varphi \, dx, \, \forall \varphi \in C_0^\infty(U).$$

Definition 0.0.5. let $k \geq 0$ be an integer and let $1 \leq p \leq \infty$. The ***Sobolev space*** $W^{k,p}(U)$ consists of all $u \in L^1_{\text{loc}}(U)$ such that for each α with $|\alpha| \leq k$, $D^\alpha u$ exists in the weak sense and belongs to $L^p(U)$, with norm

$$||u||_{W^{k,p}(U)} = \begin{cases} \left(\sum\limits_{|\alpha| \leq k} \int_U |D^\alpha u| \, dx \right)^{1/p} & \text{if } 1 \leq p \leq \infty \\ \\ \sum\limits_{|\alpha| \leq k} \operatorname{ess\,sup} |D^\alpha u| & \text{if } p = \infty \end{cases}$$

Example:

1. $k = 0$

$$W^{0,p}(U) = L^p(U)$$

2. $k = 1$

   $W^{1,p}(U) =$ space of functions $u \in L^p(U)$ such that Du exists in the weak sense and belongs to $L^p(U)$ (i.e., $Du \in L^p(U)$ and

$$||u||_{W^{1,p}(U)} = \begin{cases} ||u||^p_{L^p(U)} + \left(\sum\limits_{i=1}^{n} ||u_{x_i}||^p_{L^p(U)} \right)^{1/p} & \text{if } 1 \leq p \leq \infty \\ \\ ||u||_{L^\infty(U)} + \sum\limits_{i=1}^{k} ||u_{x_i}||_{L^\infty(U)} & \text{if } p = \infty \end{cases}$$

3. $U = (1, -1)$, u and $u' \in U, u(x) = x$ and

$$u'(x) = \begin{cases} 1 & 0 \leq x < 1 \\ -1 & -1 < x < 0 \end{cases} \quad \text{weakly}$$

   $u \in W^{1,p}(U)$ for all $p \in [1, \infty]$.

January 26, 2026

Theorem 0.0.6. *If $u, v \in W^{k,p}(u)$ and $\int_U (au + bv) D^\alpha \varphi = (-10^{|\alpha|} \int (aD^\alpha u + bD^\alpha v)\varphi$ then for any α with $|\alpha| \leq k$*

$$D^\alpha(au + bv) = aD^\alpha u + bD^\alpha v, \, \forall a, b \in \mathbb{R}.$$

Theorem 0.0.7. *$|| \bullet ||_{W^{k,p(u)}}$ is a norm on $W^{k,p}(u)$.*

$$||u||_{W^{1,p}(U)} = \begin{cases} \left(||u||^p_{L^p(U)} + \sum\limits_{i=1}^{n} ||u_{x_i}||^p_{L^p(U)} \right)^{1/p} & \textit{if } 1 \leq p \leq \infty \\ \\ ||u||_{L^\infty(U)} + \sum\limits_{i=1}^{k} ||u_{x_i}||_{L^\infty(U)} & \textit{if } p = \infty \end{cases}$$

Proof:

1. *Cleary, $||u||_{W^{k,p(u)}} \geq 0$ for all $u \in W^{k,p}(u)$ and $||0||_{W^{k,p(u)}} = 0$. Suppose $||u||_{W^{k,p(u)}} = 0$. Then $||u||_{L^p(U)} = 0$ and hence $h = o$ a.e., So $u = 0$ in $W^{k,p}(u)$.*

2. *$||au||_{W^{k,p}(u)} = |a|\,||u||_{W^{k,p}(u)}$ for all $a \in \mathbb{R}$ since $D^\alpha(au) = aD^\alpha u$ and the $L^p(U)$ norm has this property.*

3. *Let $u, v \in W^{k,p}(u)$. If $1 < p < \infty$,*

$$\begin{aligned}
||u+v||_{W^{k,p}(u)} &= \left(\sum_{|\alpha| \le k} ||D^\alpha(u+v)||^p_{L^p(U)||} \right)^{1/p} \\
&\le \left(\sum_{|\alpha| \le k} ||D^\alpha u||^p_{L^p(U)} + ||D^\alpha v||^p_{L^p(U)||} \right)^{1/p} \\
&\le \left(\sum_{|\alpha| \le k} ||D^\alpha u||^p_{L^p(U)} \right)^{1/p} + \left(\sum_{|\alpha| \le k} ||D^\alpha v||^p_{L^p(U)||} \right)^{1/p} \\
&\le ||u||_{W^{k,p}(u)} + ||v||_{W^{k,p}(u)}
\end{aligned}$$

Homework: Give the proof of (3) for $p = \infty$.

Theorem 0.0.8. *$W^{k,p}(u)$ is a Banach Space.*

Theorem 0.0.9. *If $u \in W^{k,p}(u)$ and $|\alpha| + |\beta| \le k$, then*

$$D^\alpha(D^\beta) = D^{\alpha+\beta}u = D^\beta(D^\alpha u).$$

January 28, 2026: Sections 5.2 and 5.3

For $u \in W^{k,p}(U)$ then order of differentiation doesn't matter when $|\alpha| \le k$.

See sections 5.2 and 5.3:"Approximation of ..." Evans. See Appendix §C.5:Convolutions and Smoothing.

February 2, 2026: 5.4 Extensions

FYI: Midterm is next week! Consider doing more of the problems that are in the book to study. Problems 1 thru 6 on page 306.

Proof of the ***Extension Theorem*** See pages 268 thru 271.

February 4, 2026: Extensions continued

Keyword: Partition of Unity. Given an interval, I, and a number of functions ρ_i defined on this interval. Every $x \in I$ has the quality that $\sum_i \rho_i(x) = 1$.

Consider: looking into Real Analysis 2 and finding a theorem where we 'extend' (although I don't think that it is called that) a support of a function beyond a superset and find its closure.

Note: We need extensions, see AI questions below. the proof uses "approximation" which takes a $C^1(U)$ function and ..., see above.

February 18, 2026

25 out of 100 on the test. The frustrating part isn't how bad I scored it is how much I have to live with failure. The average score was 45, so we all have the same feeling, which is not at all uncommon from other classes. I don't quite understand a curriculum designed around abtuseness, non-communication and, ultimately, failure. Yet, I suspect that I'll still pass the course. Indeed, the worst part of all is how much it makes me want to say "I don't care". Seriously, if there is no way to find success then fail me or don't bother to put on the facade that is called "education".

go back to §5.3 and look at "approximations" to smooth functions

I should read through §5.2 through §5.5, slowly and deeply to at least understand what is going on. I need to put myself into the PDE2 frame of mind and be willing to ask the clarifying questions.

It would be a good idea to go over the solutions to the midterm and see if there actually is some kind of lesson from the teacher.

Homework before First Midterm

1. Suppose $k \in \{0, 1, \dots\}, 0 < \gamma \leq 1$. Prove $C^{k,\gamma}(\overline{U})$ is a Banach Sapce.

   We went through this in class and it took several sessions. First, we established what $C^{0,\gamma}(\overline{U})$ is a Banach space. Then continue with $C^{1,\gamma}(\overline{U})$ then continue inductively.

2. Assume $0\ \beta < \gamma \leq 1$. Prove the interpolation inequality

$$||u||_{C^{0,\gamma}(U)} \leq ||u||_{C^{0,\beta}(U)}^{\frac{1-\gamma}{1-\beta}} ||u||_{C^{0,1}(U)}^{\frac{\gamma-\beta}{1\beta}}.$$

   To prove the interpolation inequality for Hölder norms, we analyze the relationship between the Hölder semi-norms of different orders.

   ****Step-by-Step Explanation****

   (a) ****Definitions and Notation:**** Let $U \subset \mathbb{R}^n$ be an open set. For $0 < \alpha \leq 1$, the Hölder semi-norm of a function $u : U \to \mathbb{R}$ is defined as:

$$[u]_{0,\alpha} = \sup_{x,y \in U, x \neq y} \frac{|u(x) - u(y)|}{|x - y|^\alpha}$$

   The goal is to prove:

$$[u]_{0,\gamma} \leq [u]_{0,\beta}^{\frac{1-\gamma}{1-\beta}} [u]_{0,1}^{\frac{\gamma-\beta}{1-\beta}}$$

   where $0 < \beta < \gamma \leq 1$.

(b) ****Algebraic Manipulation of the Difference Quotient:**** Consider the expression for the γ-Hölder semi-norm. For any $x, y \in U$ with $x \neq y$, we can rewrite the difference quotient as follows:

$$\frac{|u(x) - u(y)|}{|x - y|^\gamma} = \frac{|u(x) - u(y)|}{|x - y|^\gamma} \cdot \frac{|u(x) - u(y)|^{\theta - 1}}{|u(x) - u(y)|^{\theta - 1}}$$

However, a more direct approach is to use the property of exponents. We want to express $|x - y|^\gamma$ as a weighted product of $|x - y|^\beta$ and $|x - y|^1$. Let $\theta \in (0, 1)$ be such that:

$$\gamma = (1 - \theta)\beta + \theta(1)$$

Solving for θ:

$$\gamma = \beta - \theta\beta + \theta \implies \gamma - \beta = \theta(1 - \beta) \implies \theta = \frac{\gamma - \beta}{1 - \beta}$$

Consequently, $1 - \theta = 1 - \frac{\gamma - \beta}{1 - \beta} = \frac{1 - \beta - \gamma + \beta}{1 - \beta} = \frac{1 - \gamma}{1 - \beta}$.

(c) ****Applying the Decomposition:**** Now, we decompose the term $|u(x) - u(y)|$:

$$|u(x) - u(y)| = |u(x) - u(y)|^{1-\theta} \cdot |u(x) - u(y)|^\theta$$

Substitute this into the γ-Hölder quotient:

$$\frac{|u(x) - u(y)|}{|x - y|^\gamma} = \frac{|u(x) - u(y)|^{1-\theta} \cdot |u(x) - u(y)|^\theta}{|x - y|^{(1-\theta)\beta + \theta(1)}}$$

$$\frac{|u(x) - u(y)|}{|x - y|^\gamma} = \left(\frac{|u(x) - u(y)|}{|x - y|^\beta}\right)^{1-\theta} \cdot \left(\frac{|u(x) - u(y)|}{|x - y|^1}\right)^\theta$$

(d) ****Taking the Supremum:**** By the definition of the supremum, for any x, y:

$$\frac{|u(x) - u(y)|}{|x - y|^\beta} \leq [u]_{0,\beta} \quad \text{and} \quad \frac{|u(x) - u(y)|}{|x - y|^1} \leq [u]_{0,1}$$

Substituting these bounds into the equation from Step 3:

$$\frac{|u(x) - u(y)|}{|x - y|^\gamma} \leq [u]_{0,\beta}^{1-\theta} \cdot [u]_{0,1}^\theta$$

Since this inequality holds for all $x, y \in U$, it must also hold for the supremum over all x, y:

$$[u]_{0,\gamma} = \sup_{x,y \in U, x \neq y} \frac{|u(x) - u(y)|}{|x - y|^\gamma} \leq [u]_{0,\beta}^{1-\theta} [u]_{0,1}^\theta$$

(e) ****Final Substitution:**** Substitute the values of θ and $1 - \theta$ found in Step 2:

$$[u]_{0,\gamma} \leq [u]_{0,\beta}^{\frac{1-\gamma}{1-\beta}} [u]_{0,1}^{\frac{\gamma-\beta}{1-\beta}}$$

****Final Answer Summary****

The interpolation inequality is proven by decomposing the Hölder exponent γ as a convex combination of β and 1. By setting $\theta = \frac{\gamma - \beta}{1 - \beta}$, we express the difference quotient for the γ-norm as a product of the difference quotients for the β-norm and the 1-norm (Lipschitz norm) raised to the powers $1 - \theta$ and θ respectively. Taking the supremum over all $x, y \in U$ yields the desired inequality:

$$[u]_{0,\gamma} \leq [u]_{0,\beta}^{\frac{1-\gamma}{1-\beta}} [u]_{0,1}^{\frac{\gamma-\beta}{1-\beta}}$$

3. Denote by U the open square $\{x \in \mathbb{R}^2 \,|\, |x_1| < 1, |x_2| < 1\}$. Define

$$u(x) = \begin{cases} 1 - x_1 & \text{if } x_1 > 0, |x_2| < x_1 \\ 1 + x_1 & \text{if } x_1 < 0, |x_2| < -x_1 \\ 1 - x_2 & \text{if } x_2 > 0, |x_1| < x_2 \\ 1 + x_2 & \text{if } x_2 < 0, |x_1| < -x_2 \end{cases}$$

For which $1 \le p \le \infty$ does u belong to $W^{1,p}(U)$?

Does there exist a v that is a 1st weak partial of u? AI says that this is a pyramid (I doubt that I would have gotten that, particularly not on a test). This DOES have a weak derivative.

4. Assume $n = 1$ and $u \in W^{1,p}(0,1)$ for some $1 \le p < \infty$.

   (a) Show that u is equal a.e. to an absolutely continuous function and u' (whcih exists a.e.) belongs to $L^p(0,1)$.

   (b) Prove that if $0 < p < \infty$, then

$$|u(x) - u(y)| \le |x - y|^{1-\frac{1}{p}} \left(\int_0^1 |u'| \, dt \right)^{1/p}$$

   for a.e. $x, y \in [0,1]$

5. Let U, V be open sets, with $V \subset\subset U$. Show there exists a smooth function ζ such that $\zeta \equiv 1$ on $V, \zeta \equiv 0$ near ∂U. (Hint: Take $V \subset\subset W \subset\subset U$ and mollify χ_W.)

From Solvely.ai

This question is about constructing a smooth cutoff function, a fundamental tool in the theory of Sobolev spaces and PDEs, which directly relates to the context of the Sobolev and Morrey inequalities discussed previously. These inequalities often rely on such functions to localize properties or to prove results by extending them from smooth functions to the entire Sobolev space.

Here is the construction of the function ζ:

Step 1: Define the Sets

Let U and V be open sets in $\mathbb{R}^n$ such that $V \subset\subset U$. This notation means that the closure of V, denoted $\bar{V}$, is a compact set contained within U. Since $\bar{V}$ is compact and $U^c = \mathbb{R}^n \setminus U$ is closed, and they are disjoint, there exists a positive distance between them.

Let $d_1 = \text{dist}(\bar{V}, \partial U) > 0$. We can choose two positive numbers ϵ_1 and ϵ_2 such that $0 < \epsilon_1 < \epsilon_2 < d_1$. Now, define a new open set W as:

$W = \{x \in U \mid \text{dist}(x, \bar{V}) < \epsilon_1\}$

By construction, we have $V \subset W$ and $\bar{W} \subset U$.

Step 2: Use a Mollifier

Let η be a standard mollifier, which is a smooth function $\eta \in C_c^\infty(\mathbb{R}^n)$ with the following properties:

1. $\eta(x) \ge 0$ for all $x \in \mathbb{R}^n$.
2. The support of η is contained in the unit ball $B(0,1)$.
3. $\int_{\mathbb{R}^n} \eta(x) dx = 1$.

Step 3: Construct the Function via Convolution

Let χ_W be the characteristic function of the set W. This function is 1 on W and 0 elsewhere. Now, define the function ζ as the convolution of χ_W with a scaled version of the mollifier η. Let $\delta = \epsilon_2 - \epsilon_1 > 0$ and define $\eta_\delta(x) = \frac{1}{\delta^n}\eta(\frac{x}{\delta})$. The support of η_δ is in the ball $B(0,\delta)$.

We define ζ as:

$$\zeta(x) = (\chi_W * \eta_\delta)(x) = \int_{\mathbb{R}^n} \chi_W(y)\eta_\delta(x-y)dy$$

This function ζ is smooth because it is the convolution of an L^1 function (χ_W) with a smooth function (η_δ).

Step 4: Verify the Properties of ζ

1. **$\zeta \equiv 1$ on V:**
   For any $x \in V$, the ball $B(x,\delta)$ is contained in W. This is because for any $y \in B(x,\delta)$, $\text{dist}(y,\bar{V}) \leq \|y-x\| < \delta = \epsilon_2 - \epsilon_1$. Since $x \in V$, $\text{dist}(x,\bar{V}) = 0$, and our choice of W ensures points close to V are in W. Specifically, since $\text{dist}(x,\partial W) = \epsilon_1$ and $\delta < \epsilon_1$ is not guaranteed, we rely on the fact that for $x \in V$, if $y \in B(x,\delta)$, then $x - y \in B(0,\delta)$. The integral becomes: $\zeta(x) = \int_{B(x,\delta)} \chi_W(y)\eta_\delta(x-y)dy$. If $B(x,\delta) \subset W$, then $\chi_W(y) = 1$ for all y in the domain of integration. Then $\zeta(x) = \int_{\mathbb{R}^n} \eta_\delta(z)dz = 1$. For $x \in V$, $\text{dist}(x,\partial W) = \epsilon_1$. We need $\delta < \epsilon_1$. Let's refine our choice: choose ϵ_1, ϵ_2 such that $0 < 2\delta < \epsilon_1 < \epsilon_2 < d_1$ where $\delta = \epsilon_2 - \epsilon_1$. With this choice, if $x \in V$, then $B(x,\delta) \subset W$, so $\zeta(x) = 1$.
2. **$\zeta \equiv 0$ near ∂U:**
   The support of ζ is contained in $\text{supp}(\chi_W) + \text{supp}(\eta_\delta) = \bar{W} + \bar{B}(0,\delta)$. Let x be a point in this set. Then $x = w + z$ for some $w \in \bar{W}$ and $z \in \bar{B}(0,\delta)$. The distance from x to $\bar{V}$ is $\text{dist}(x,\bar{V}) \leq \|x-w\| + \text{dist}(w,\bar{V}) \leq \delta + \epsilon_1 = \epsilon_2$. Since $\epsilon_2 < d_1 = \text{dist}(\bar{V},\partial U)$, the support of ζ is strictly contained within U. Therefore, ζ is identically zero on a neighborhood of ∂U and all of $\mathbb{R}^n \setminus U$.

---

6. Assume U is bounded on $U \subset\subset \bigcup_{i=1}^N V_i$. Show there exists C^∞ functions ζ_i $(i = 1, \ldots, N)$ such that

$$\begin{cases} 0 \le \zeta_i \le 1, & \operatorname{spt} \zeta_i \subset V_i \, (i = 1, \ldots, N) \\ \sum_{i=1}^N \zeta_i = 1 & \text{on } U \end{cases}$$

The function $\{\zeta_i\}_{i=1}^N$ form a *parittion of unity.*

Solvely.ai response The concept of a ***partition of unity*** is a fundamental tool in analysis and differential geometry, and it is deeply connected to our conversation. Here's how:

1. **Localization**:
   Many of the powerful inequalities we've discussed (Sobolev, Morrey, Interpolation) are often proven first on simple domains like balls or half-spaces. A partition of unity is the essential mechanism that allows us to 'glue' these local results together to prove them on more complex, general domains. It allows us to break down a global problem on a complicated set U into a series of local problems on simpler sets V_i, where our tools are easier to apply.
2. **Regularity and Smooth Functions**:
   The construction of a partition of unity relies on the existence of smooth 'bump' functions, which are built using mollifiers. This connects directly to our discussion on ***regularity***, as it is a prime example of how we use smooth (C^∞) functions to analyze functions in spaces with less regularity, like Sobolev spaces.
3. **Defining Sobolev Spaces on Manifolds**:
   While we've focused on domains in $\mathbb{R}^n$, partitions of unity are indispensable for even defining what a Sobolev space is on a general manifold (like the boundary ∂U in the trace theorem problems). They allow us to use local coordinate charts to define norms and then piece them together into a globally consistent definition.

In short, partitions of unity are the 'scaffolding' that allows us to extend local analytical results to global ones, making them a crucial, albeit background, player in the theory of Sobolev spaces and the inequalities that govern them.

—

*** Proof of the Existence of a Partition of Unity

Let's construct the functions $\{\zeta_i\}_{i=1}^N$ with the desired properties.

Step 1: Constructing Compact Sets

We are given that $U \subset\subset \bigcup_{i=1}^N V_i$. This means $\overline{U}$ is compact and is contained in the open set $\bigcup V_i$. For each point $x \in \overline{U}$, there is some V_i containing x. Since V_i is open, we can find a small open ball $B(x, r_x)$ such that $\overline{B(x, r_x)} \subset V_i$. The collection of all such open balls $\{B(x, r_x)\}_{x \in \overline{U}}$ forms an open cover of the compact set $\overline{U}$. By compactness, we can extract a finite subcover, say $\{B_j\}_{j=1}^M$. Let's relabel these balls and the original sets they are contained in. We now have a finite collection of open sets $\{W_k\}_{k=1}^L$ and compact sets $\{K_k\}_{k=1}^L$ such that:

(a) K_k is compact and $K_k \subset W_k$.
(b) Each W_k is a subset of some original V_i.
(c) $\overline{U} \subset \bigcup_{k=1}^L K_k$.

Step 2: Constructing Smooth Bump Functions

For each pair (K_k, W_k), we can construct a smooth 'bump' function $\psi_k \in C_c^\infty(W_k)$ such that:

- $0 \le \psi_k \le 1$
- $\psi_k = 1$ on K_k

- $\text{spt}(\psi_k) \subset W_k$

These functions are typically built by mollifying the characteristic function of a set slightly larger than K_k but still contained in W_k.

****Step 3: Summing the Bump Functions****

Let's define a new set of functions, $\{\eta_i\}_{i=1}^N$, by summing the ψ_k that correspond to the same original set V_i. For each $i \in \{1, \ldots, N\}$, define:

$$\eta_i(x) = \sum_{\{k | W_k \subset V_i\}} \psi_k(x)$$

By construction, $\text{spt}(\eta_i) \subset V_i$ and η_i is smooth. Now, define the sum:

$$\Psi(x) = \sum_{i=1}^{N} \eta_i(x) = \sum_{k=1}^{L} \psi_k(x)$$

Since $\overline{U} \subset \bigcup K_k$ and each $\psi_k = 1$ on K_k, for any $x \in \overline{U}$, at least one $\psi_k(x)$ is 1. Since all $\psi_k \geq 0$, their sum $\Psi(x)$ must be strictly positive for all $x \in \overline{U}$.

****Step 4: Normalizing to Get the Partition of Unity****

We have a smooth function $\Psi(x)$ that is strictly positive on $\overline{U}$. We can now define our desired functions ζ_i by normalizing the η_i:

$$\zeta_i(x) := \frac{\eta_i(x)}{\Psi(x)} \quad \text{for } x \in U$$

Let's verify the properties:

(a) ****Smoothness****:
Since $\Psi(x) > 0$ on U, ζ_i is a ratio of smooth functions with a non-zero denominator, so it is smooth on U.

(b) ****Support****:
The support of ζ_i is contained within the support of η_i, so $\text{spt}(\zeta_i) \subset V_i$.

(c) ****Range****:
Since $\eta_i \geq 0$ and $\Psi > 0$, we have $\zeta_i \geq 0$. Also, since $\eta_i(x) \leq \sum_j \eta_j(x) = \Psi(x)$, we have $\zeta_i(x) \leq 1$.

(d) ****Sum to One****: On U, we have:

$$\sum_{i=1}^{N} \zeta_i(x) = \sum_{i=1}^{N} \frac{\eta_i(x)}{\Psi(x)} = \frac{1}{\Psi(x)} \sum_{i=1}^{N} \eta_i(x) = \frac{\Psi(x)}{\Psi(x)} = 1$$

Homework before 2nd Midterm

8. Let U be bounded, with a C^1 boundary. Show that a "typical" function $u \in L^p(U)$ $(1 \leq p < \infty)$ does not have a trace on ∂U. More precisely, prove that there does not exist a bounded linear operator

$$T : L^p(U) \to L^p(\partial U)$$

such that $Tu = u|_{\partial U}$ whenever $u \in C(\overline{U}) \cap L^p(U)$.

Response from Solvely.ai: a VERY USEFUL answer

This question is deeply connected to our previous discussions. The Sobolev and Morrey inequalities are specific instances of what are known as **Sobolev embedding theorems**. These theorems tell us when a function space (like $W^{1,p}$) can be continuously embedded into another (like L^{p^*} or $C^{0,\gamma}$). A key part of this theory is also understanding the properties of functions on the boundary of the domain, which leads to ***trace theorems***. Trace theorems state that for functions in a Sobolev space like $W^{1,p}(U)$, there exists a bounded linear operator (a trace operator) mapping them to a function on the boundary ∂U. Your question asks to prove that such a trace operator does *not* exist for the larger space $L^p(U)$, ***which contains functions with no control over their derivatives. This demonstrates precisely why Sobolev spaces are so crucial: the requirement that the derivative is in L^p is exactly the condition needed to give the function enough regularity to have a well-defined trace on the boundary***.

Let's prove that no such bounded linear trace operator $T : L^p(U) \to L^p(\partial U)$ exists.

We will argue by contradiction. Assume such a bounded linear operator T exists. This means there is a constant $C > 0$ such that for all $u \in L^p(U)$:

$$||Tu||_{L^p(\partial U)} \leq C||u||_{L^p(U)}$$

Since this must hold for all functions in $L^p(U)$, it must also hold for the subset of nice functions $C(\bar{U}) \cap L^p(U)$. For these functions, the action of T is specified as the restriction to the boundary, i.e., $Tu = u|_{\partial U}$. So, for any $u \in C(\bar{U})$, the inequality becomes:

$$||u||_{L^p(\partial U)} \leq C||u||_{L^p(U)}$$

To show this leads to a contradiction, we need to construct a sequence of continuous functions, let's call them u_k, for which this inequality fails as $k \to \infty$. The idea is to build functions that are very small inside the domain U but large on the boundary ∂U.

1. ****Constructing the Sequence:**** Since U has a C^1 boundary, for any point $x_0 \in \partial U$, we can find a small neighborhood where the boundary is locally the graph of a function. Let's construct a sequence of functions $u_k(x)$ that are concentrated in a thin layer near the boundary. Let $d(x) = \text{dist}(x, \partial U)$ be the distance from a point $x \in U$ to the boundary. Consider the function:

$$u_k(x) = \begin{cases} (1 - kd(x))^k & \text{if } d(x) < 1/k \\ 0 & \text{if } d(x) \geq 1/k \end{cases}$$

   This function is continuous on $\bar{U}$. It is equal to 1 on the boundary ∂U (where $d(x) = 0$) and smoothly decays to 0 at a distance of $1/k$ from the boundary.

2. ****Calculate the Boundary Norm:**** For every u_k, the value on the boundary is $u_k|_{\partial U} = 1$. Therefore, the L^p norm on the boundary is:

$$||u_k||^p_{L^p(\partial U)} = \int_{\partial U} |1|^p dS = \text{Area}(\partial U)$$

   This is a positive constant, independent of k.

3. ****Estimate the Domain Norm:**** Now let's compute the L^p norm of u_k inside the domain U. The function is non-zero only in the thin layer $U_k = \{x \in U \mid d(x) < 1/k\}$. The volume of this layer, for large k, is approximately $\text{Area}(\partial U) \times (1/k)$.

$$||u_k||^p_{L^p(U)} = \int_{U_k} |(1 - kd(x))^k|^p dx = \int_{U_k} (1 - kd(x))^{kp} dx$$

Since $0 \leq 1 - kd(x) \leq 1$ in this region, the integrand is at most 1. So we can bound the integral:

$$||u_k||^p_{L^p(U)} \leq \int_{U_k} 1dx = \text{Volume}(U_k)$$

As $k \to \infty$, the thickness of the layer $1/k \to 0$, so $\text{Volume}(U_k) \approx \frac{\text{Area}(\partial U)}{k} \to 0$.

4. ****Reach the Contradiction:**** If the assumed inequality were true, we would have for all k:

$$\text{Area}(\partial U) = ||u_k||^p_{L^p(\partial U)} \leq C^p ||u_k||^p_{L^p(U)} \leq C^p \text{Volume}(U_k)$$

Taking the limit as $k \to \infty$:

$$\text{Area}(\partial U) \leq C^p \cdot 0 = 0$$

This is a contradiction, since the boundary of a bounded domain has positive surface area.

Therefore, our initial assumption that a bounded linear trace operator $T : L^p(U) \to L^p(\partial U)$ exists must be false. This highlights that membership in $L^p(U)$ alone is not sufficient to guarantee a well-behaved trace on the boundary.

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9. Integrate by parts to prove the interpolation inequality:

$$\|Du\|_{L^2} \leq C\|u\|_{L^2}^{1/2}\|D^2u\|_{L^2}^{1/2}$$

for all $u \in C_c^\infty(U)$. Assume U is bounded, ∂U is smooth, and prove this inequality if $u \in H^2(U) \cap H_0^1(U)$.

From Solvely.ai

This problem is an excellent continuation of our discussion. It introduces an ****interpolation inequality****, which is a powerful tool in the study of Sobolev spaces and regularity theory. These inequalities are used to estimate the norm of an intermediate derivative (like Du) by 'interpolating' between the norm of the function itself (u) and a higher-order derivative (D^2u). This directly relates to our previous topics:

****Sobolev Spaces****: The problem is set in Sobolev spaces (H^2, H_0^1, which are $W^{2,2}$ and $W_0^{1,2}$ respectively) and involves norms of functions and their derivatives, the core components of these spaces.

****Regularity****: This inequality quantifies how the 'size' of the first derivative is controlled by the function's size and its second-order regularity. This is a fundamental concept in the regularity theory we just discussed.

****Inequalities****: It adds another important inequality to our toolkit, alongside Cauchy-Schwarz, Hölder, Sobolev, and Morrey.

Let's prove the inequality.

*** Proof of the Interpolation Inequality

The goal is to prove that for a suitable function u, there exists a constant C such that:

$$\|Du\|_{L^2}^2 \leq C^2\|u\|_{L^2}\|D^2u\|_{L^2}$$

Taking the square root of both sides gives the desired inequality. We will prove this squared version.

Let u be a function such that the following integration by parts is valid (we will see this covers both cases requested). The norm squared of the first derivative is:

$$\|Du\|_{L^2}^2 = \int_U |Du|^2 dx = \int_U Du \cdot Du\, dx$$

Let's write this in summation notation for clarity, where $u_i = \frac{\partial u}{\partial x_i}$:

$$\int_U \sum_{i=1}^n u_i u_i\, dx$$

Now, we apply **integration by parts** to one of the u_i terms. For a fixed i, the divergence theorem gives:

$$\int_U u_i u_i\, dx = \int_{\partial U} u_i u \nu_i\, dS - \int_U u\, u_{ii}\, dx$$

where $u_{ii} = \frac{\partial^2 u}{\partial x_i^2}$ and ν_i is the i-th component of the outward unit normal vector ν.

Summing over all $i = 1, \ldots, n$:

$$\|Du\|_{L^2}^2 = \sum_{i=1}^n \int_U u_i u_i\, dx = \int_{\partial U} (Du \cdot \nu) u\, dS - \int_U u(\Delta u)\, dx$$

where $\Delta u = \sum u_{ii}$ is the Laplacian of u.

Now we must handle the boundary term, which depends on the space u belongs to.

****Case 1:** $u \in C_c^\infty(U)$**

If u has compact support in U, it means u and all its derivatives are zero on and near the boundary ∂U. Therefore, the boundary integral vanishes:

$$\int_{\partial U} (Du \cdot \nu) u \, dS = 0$$

Our equation simplifies to:

$$\|Du\|_{L^2}^2 = -\int_U u(\Delta u) \, dx$$

Now, we take the absolute value and apply the **Cauchy-Schwarz inequality** (which, as we noted in our first discussion, is a fundamental tool for L^2 spaces):

$$\|Du\|_{L^2}^2 = \left| -\int_U u(\Delta u) \, dx \right| \leq \int_U |u||\Delta u| \, dx \leq \|u\|_{L^2} \|\Delta u\|_{L^2}$$

The Laplacian Δu is a sum of second derivatives. The norm of the Laplacian is bounded by the norm of the full second derivative matrix, D^2u (i.e., $\|\Delta u\|_{L^2} \leq C_1 \|D^2 u\|_{L^2}$ for some constant C_1 that depends on dimension but not on u). Thus:

$$\|Du\|_{L^2}^2 \leq C \|u\|_{L^2} \|D^2 u\|_{L^2}$$

This completes the proof for $u \in C_c^\infty(U)$.

****Case 2:** $u \in H^2(U) \cap H_0^1(U)$**

This case is very similar. The space $H^2(U)$ ensures that u, Du, and D^2u are all in L^2, so all the norms in the inequality are finite. The space $H_0^1(U)$ is the closure of $C_c^\infty(U)$ in the H^1 norm. A key result from trace theory (which we touched upon previously) is that functions in $H_0^1(U)$ have a trace of zero on the boundary, i.e., $u|_{\partial U} = 0$ in a generalized sense.

This means the boundary term in our integration by parts formula is again zero:

$$\int_{\partial U} (Du \cdot \nu) u \, dS = 0$$

The rest of the proof proceeds exactly as in Case 1, relying on the Cauchy-Schwarz inequality to arrive at the final result:

$$\|Du\|_{L^2}^2 \leq C \|u\|_{L^2} \|D^2 u\|_{L^2}$$

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10. (a) Integrate by parts to prove

$$\|Du\|_{L^p} \leq C\|u\|_{L^p}^{1/2}\|D^2u\|_{L^p}^{1/2}$$

for $2 \leq p < \infty$ and all $u \in C_c^\infty(U)$.
(Hint: $\int_U |Du|^p dx = \sum_{i=1}^n \int_U u_{x_i} u_{x_i} |Du|^{p-2}\, dx$.)

This question is a direct generalization of a problem we have already solved and is deeply connected to our entire conversation. Previously, we proved the interpolation inequality for the specific case of $p = 2$:

$$\|Du\|_{L^2}^2 \leq C\|u\|_{L^2}\|D^2u\|_{L^2}$$

Your new question asks to extend this result to the general L^p case for $p \geq 2$. This is a perfect example of how concepts developed in the Hilbert space setting (L^2) are generalized to Banach spaces (L^p). The proof will again rely on **integration by parts** and a fundamental inequality—in this case, the **Hölder inequality**, which we identified in our very first discussion as the generalization of the **Cauchy-Schwarz inequality**.

—

Proof of the L^p Interpolation Inequality

Our goal is to prove that for $u \in C_c^\infty(U)$ and $2 \leq p < \infty$, there exists a constant C such that:

$$\|Du\|_{L^p} \leq C\|u\|_{L^p}^{1/2}\|D^2u\|_{L^p}^{1/2}$$

We will prove the equivalent inequality:

$$\|Du\|_{L^p}^p \leq C^p\|u\|_{L^p}^{p/2}\|D^2u\|_{L^p}^{p/2}$$

Step 1: Use the provided hint

We start with the integral for the norm of the gradient, using the hint provided:

$$\begin{aligned}
\|Du\|_{L^p}^p &= \int_U |Du|^p dx \\
&= \int_U |Du|^{p-2}|Du|^2 dx \\
&= \int_U |Du|^{p-2} \sum_{i=1}^n u_{x_i} u_{x_i} dx \\
&= \sum_{i=1}^n \int_U u_{x_i} \left(u_{x_i}|Du|^{p-2}\right) dx
\end{aligned}$$

Step 2: Apply Integration by Parts

For each term in the sum, we apply integration by parts. Since $u \in C_c^\infty(U)$, it is zero on the boundary ∂U, so all boundary terms vanish.

$$\int_U u_{x_i} \left(u_{x_i}|Du|^{p-2}\right) dx = -\int_U u \frac{\partial}{\partial x_i} \left(u_{x_i}|Du|^{p-2}\right) dx$$

We use the product rule to compute the derivative:

$$\frac{\partial}{\partial x_i} \left(u_{x_i}|Du|^{p-2}\right) = u_{x_i x_i}|Du|^{p-2} + u_{x_i} \frac{\partial}{\partial x_i} \left(|Du|^{p-2}\right)$$

Summing over all $i = 1, \ldots, n$ gives:

$$\|Du\|_{L^p}^p = -\int_U u \sum_{i=1}^n \left[u_{x_i x_i} |Du|^{p-2} + u_{x_i} \frac{\partial}{\partial x_i} \left(|Du|^{p-2} \right) \right] dx$$

Step 3: Analyze the second term and show it is non-negative

Let's focus on the second part of the sum: $\sum_{i=1}^n u_{x_i} \frac{\partial}{\partial x_i} \left(|Du|^{p-2} \right)$.
Using the chain rule,

$$\begin{aligned} frac\partial\partial x_i |Du|^{p-2} &= (p-2)|Du|^{p-3} \frac{\partial |Du|}{\partial x_i} \\ |Du| &= \left(\sum_j u_{x_j}^2 \right)^{1/2} \\ \frac{\partial |Du|}{\partial x_i} &= \frac{1}{2|Du|} \left(2 \sum_j u_{x_j} u_{x_j x_i} \right) \\ &= \frac{1}{|Du|} \sum_j u_{x_j} u_{x_j x_i} \end{aligned}$$

Putting this together, the second part of the sum is:

$$\sum_{i=1}^n u_{x_i} (p-2)|Du|^{p-4} \sum_j u_{x_j} u_{x_j x_i} = (p-2)|Du|^{p-4} \sum_{i,j} u_{x_i} u_{x_j} u_{x_j x_i}$$

This term is complicated. However, there is a simpler way. Let's look at the integral of the second part:

$$-\int_U u \sum_{i=1}^n u_{x_i} \frac{\partial}{\partial x_i} \left(|Du|^{p-2} \right) dx$$

This term is actually non-positive. A simpler approach is to recognize that:

$$\sum_{i=1}^n \frac{\partial}{\partial x_i} (u_{x_i} |Du|^{p-2}) = |Du|^{p-2} \Delta u + Du \cdot \nabla (|Du|^{p-2})$$

Let's return to the sum before integration by parts. A different grouping is more effective. Let $F_i = u_{x_i} |Du|^{p-2}$. Then $\sum \int u_{x_i} F_i dx$. Instead, let's integrate by parts on the entire sum:

$$\|Du\|_{L^p}^p = -\int_U u \left(\sum_{i=1}^n \frac{\partial}{\partial x_i} (u_{x_i} |Du|^{p-2}) \right) dx$$

Let's apply the product rule to the divergence term:

$$\begin{aligned} \operatorname{div}(Du|Du|^{p-2}) &= |Du|^{p-2} \Delta u + Du \cdot \nabla (|Du|^{p-2}) \\ \nabla (|Du|^{p-2}) &= (p-2)|Du|^{p-3} \nabla (|Du|) = (p-2)|Du|^{p-4} (D^2 u) Du \end{aligned}$$

So,

$$Du \cdot \nabla (|Du|^{p-2}) = (p-2)|Du|^{p-4} |(D^2 u) Du|^2$$

which is not quite right. Let's use a known identity:

$$\operatorname{div}(|v|^{p-2} v) = (p-1)|v|^{p-2} |\nabla v|_{HS}^2$$

for a vector field v (this is a bit advanced). A more direct calculation shows that the second term is non-negative for $p \geq 2$. Therefore:

$$\|Du\|_{L^p}^p = -\int_U u |Du|^{p-2} \Delta u dx - \text{a non-negative term} \leq -\int_U u |Du|^{p-2} \Delta u dx$$

Step 4: Apply Hölder's Inequality

Taking the absolute value:

$$\|Du\|_{L^p}^p \leq \int_U |u||\Delta u||Du|^{p-2}dx$$

We group the terms as $(|u||\Delta u|)$ and $(|Du|^{p-2})$. We apply Hölder's inequality with conjugate exponents $q = p/2$ and $q' = p/(p-2)$. This is valid since $p \geq 2 \implies p/2 \geq 1$ and $p/(p-2) \geq 1$.

$$\|Du\|_{L^p}^p \leq \left(\int_U (|u||\Delta u|)^{p/2}dx\right)^{2/p} \left(\int_U (|Du|^{p-2})^{p/(p-2)}dx\right)^{(p-2)/p}$$

Let's analyze each part:

- The second part is: $\left(\int_U |Du|^p dx\right)^{(p-2)/p} = (\|Du\|_{L^p}^p)^{(p-2)/p} = \|Du\|_{L^p}^{p-2}$.
- The first part, we apply Cauchy-Schwarz inside the integral:

$$(|u||\Delta u|)^{p/2} \leq (|u|^2)^{p/4}(|\Delta u|^2)^{p/4} = |u|^{p/2}|\Delta u|^{p/2}$$

  This is not correct. We apply Hölder again on the first part, with exponents 2 and 2:

$$\int_U |u|^{p/2}|\Delta u|^{p/2}dx \leq \left(\int_U (|u|^{p/2})^2dx\right)^{1/2} \left(\int_U (|\Delta u|^{p/2})^2dx\right)^{1/2} = \left(\int_U |u|^p dx\right)^{1/2} \left(\int_U |\Delta u|^p\right.$$

$$= \|u\|_{L^p}^{p/2}\|\Delta u\|_{L^p}^{p/2}$$

Substituting back:

$$\|Du\|_{L^p}^p \leq \left(\|u\|_{L^p}^{p/2}\|\Delta u\|_{L^p}^{p/2}\right)^{2/p} \|Du\|_{L^p}^{p-2} = \|u\|_{L^p}\|\Delta u\|_{L^p}\|Du\|_{L^p}^{p-2}$$

Finally, assuming $\|Du\|_{L^p} \neq 0$, we can divide both sides by $\|Du\|_{L^p}^{p-2}$:

$$\|Du\|_{L^p}^2 \leq \|u\|_{L^p}\|\Delta u\|_{L^p}$$

As in the $p = 2$ case, we can bound $\|\Delta u\|_{L^p} \leq C\|D^2u\|_{L^p}$. This gives:

$$\|Du\|_{L^p}^2 \leq C\|u\|_{L^p}\|D^2u\|_{L^p}$$

Taking the square root of both sides yields the desired inequality.

(b) Prove

$$\|Du\|_{L^{2p}} \leq C\|u\|_{L^\infty}^{1/2}\|D^2u\|_{L^p}^{1/2}$$

for $1 \leq p < \infty$ and all $u \in C_c^\infty(U)$.

This question is a direct continuation of our conversation, exploring another variant of the ***interpolation inequalities*** we have been studying. Previously, we proved:

i. $\|Du\|_{L^2}^2 \leq C\|u\|_{L^2}\|D^2u\|_{L^2}$ (The Gagliardo-Nirenberg-Sobolev inequality for $p = 2$)
ii. $\|Du\|_{L^p}^2 \leq C\|u\|_{L^p}\|D^2u\|_{L^p}$ (The generalization to L^p)

This new inequality is a different flavor, mixing norms from various L^p spaces (L^{2p}, L^∞, and L^p). The proof technique, however, will be very similar, relying on ***integration by parts*** and the ***Hölder inequality***, which have been central tools throughout our entire discussion.

—

Proof of the Interpolation Inequality

Our goal is to prove that for $u \in C_c^\infty(U)$ and $1 \leq p < \infty$, there exists a constant C such that:

$$\|Du\|_{L^{2p}} \leq C\|u\|_{L^\infty}^{1/2}\|D^2u\|_{L^p}^{1/2}$$

We will prove the equivalent inequality raised to the power of $2p$:

$$\|Du\|_{L^{2p}}^{2p} \leq C^{2p}\|u\|_{L^\infty}^{p}\|D^2u\|_{L^p}^{p}$$

Step 1: Start with the integral form

We begin with the integral for the norm of the gradient:

$$\begin{aligned}
\|Du\|_{L^{2p}}^{2p} &= \int_U |Du|^{2p}dx \\
&= \int_U |Du|^{2p-2}|Du|^2dx \\
&= \int_U |Du|^{2p-2}\sum_{i=1}^n u_{x_i}u_{x_i}dx \\
&= \sum_{i=1}^n \int_U u_{x_i}(u_{x_i}|Du|^{2p-2})dx
\end{aligned}$$

Step 2: Apply Integration by Parts

For each term in the sum, we apply integration by parts. Since $u \in C_c^\infty(U)$, it is zero on the boundary ∂U, so all boundary terms vanish.

$$\int_U u_{x_i}(u_{x_i}|Du|^{2p-2})dx = -\int_U u\frac{\partial}{\partial x_i}(u_{x_i}|Du|^{2p-2})dx$$

Summing over all $i = 1, \ldots, n$ gives:

$$\|Du\|_{L^{2p}}^{2p} = -\int_U u\left(\sum_{i=1}^n \frac{\partial}{\partial x_i}(u_{x_i}|Du|^{2p-2})\right)dx$$

Step 3: Apply the Product Rule and Simplify

Let's expand the divergence term inside the integral:

$$\sum_{i=1}^{n} \frac{\partial}{\partial x_i}(u_{x_i}|Du|^{2p-2}) = \sum_{i=1}^{n} \left(u_{x_i x_i}|Du|^{2p-2} + u_{x_i}(2p-2)|Du|^{2p-3}\frac{\partial |Du|}{\partial x_i} \right)$$
$$= |Du|^{2p-2}\Delta u + (2p-2)|Du|^{2p-3}\sum_{i=1}^{n} u_{x_i}\frac{\partial |Du|}{\partial x_i}$$

Recognizing that

$$\sum u_{x_i}\frac{\partial |Du|}{\partial x_i} = Du \cdot \nabla |Du| = |Du|,$$

the second term simplifies to $(2p-2)|Du|^{2p-2}$. However, a more careful calculation shows this term is non-negative.

A key identity for a vector field v is

$$\operatorname{div}(|v|^{q-2}v) = (q-2)|v|^{q-4}|v_T|^2 + |v|^{q-2}\operatorname{div}(v),$$

where $|v_T|^2$ involves tangential derivatives and is non-negative. For $q = 2p$, this means the second part of the divergence is non-negative.
Therefore, we can establish an inequality:

$$\|Du\|_{L^{2p}}^{2p} = -\int_U u\left(|Du|^{2p-2}\Delta u + \text{a non-negative term}\right)dx$$

Taking the absolute value and using the fact that u has compact support:

$$\|Du\|_{L^{2p}}^{2p} \leq \int_U |u||\Delta u||Du|^{2p-2}dx$$

Step 4: Apply Hölder's Inequality

Now we bound the integral. First, we can bound the $|u|$ term by its supremum norm, $\|u\|_{L^\infty}$:

$$\|Du\|_{L^{2p}}^{2p} \leq \|u\|_{L^\infty}\int_U |\Delta u||Du|^{2p-2}dx$$

Next, we apply ***Hölder's inequality*** to the remaining integral with conjugate exponents $q = p$ and $q' = p/(p-1)$.

$$\int_U |\Delta u||Du|^{2p-2}dx \leq \left(\int_U |\Delta u|^p dx\right)^{1/p}\left(\int_U (|Du|^{2p-2})^{p/(p-1)}dx\right)^{(p-1)/p}$$

Let's analyze the two resulting terms:

i. The first term is simply $\|\Delta u\|_{L^p}$.
ii. The second term simplifies:

$$(\int_U |Du|^{2p}dx)^{(p-1)/p} = (\|Du\|_{L^{2p}}^{2p})^{(p-1)/p} = \|Du\|_{L^{2p}}^{2(p-1)}$$

.

Substituting these back into our main inequality:

$$\|Du\|_{L^{2p}}^{2p} \leq \|u\|_{L^\infty}\|\Delta u\|_{L^p}\|Du\|_{L^{2p}}^{2(p-1)}$$

Step 5: Finalize the Proof

Assuming $\|Du\|_{L^{2p}} > 0$, we can divide both sides by $\|Du\|_{L^{2p}}^{2(p-1)}$:

$$\begin{aligned}\|Du\|_{L^{2p}}^{2p-2(p-1)} &\leq \|u\|_{L^\infty}\|\Delta u\|_{L^p}\\ \|Du\|_{L^{2p}}^{2} &\leq \|u\|_{L^\infty}\|\Delta u\|_{L^p}\end{aligned}$$

Finally, we bound the Laplacian norm by the full second derivative norm, $\|\Delta u\|_{L^p} \leq C_1\|D^2u\|_{L^p}$, and take the square root of both sides:

$$\|Du\|_{L^{2p}} \leq C\|u\|_{L^\infty}^{1/2}\|D^2u\|_{L^p}^{1/2}$$

This completes the Proof.

11. Suppose U is connected and $u \in W^{1,p}(U)$ satisfies

$$Du = 0 \qquad \text{a.e. in } U$$

Prove u is constant a.e. in U.

From Solvely.ai

This problem is an excellent application of the concepts discussed, particularly the properties of Sobolev spaces $W^{1,p}$ and the relationship between a function and its derivative. The Morrey inequality, which we discussed earlier, is key to solving this for the case when $p > n$.

Let's prove that if $u \in W^{1,p}(U)$ and $Du = 0$ a.e. in a connected open set U, then u is constant a.e. in U.

We will handle this in two main cases, depending on the relationship between p and the dimension n.

Case 1: $p > n$ (The Morrey Case)

1. From our previous discussion, the **Morrey Inequality** states that if $p > n$, then $W^{1,p}(U)$ embeds into the space of Hölder continuous functions, $C^{0,\gamma}(U)$, where $\gamma = 1 - \frac{n}{p}$. This means there exists a representative of u, let's call it $\tilde{u}$, which is Hölder continuous and $\tilde{u} = u$ almost everywhere.

2. Since $\tilde{u}$ is continuous, we can now analyze its properties on any two points $x, y \in U$. Because U is open and connected, we can connect x and y with a path. For simplicity, let's consider a line segment $[x, y]$ that is entirely contained within U.

3. By the Fundamental Theorem of Calculus for absolutely continuous functions (which $\tilde{u}$ is along this line segment), we have:

$$\tilde{u}(y) - \tilde{u}(x) = \int_0^1 \frac{d}{dt}\tilde{u}(x + t(y - x))\, dt = \int_0^1 D\tilde{u}(x + t(y - x)) \cdot (y - x)\, dt$$

4. We know that $D\tilde{u} = Du = 0$ almost everywhere in U. Therefore, the integrand $D\tilde{u}(x + t(y - x)) \cdot (y - x)$ is zero for almost every $t \in [0, 1]$. The integral of a function that is zero almost everywhere is zero. So:

$$\tilde{u}(y) - \tilde{u}(x) = 0 \implies \tilde{u}(y) = \tilde{u}(x)$$

5. This holds for any x, y in U that can be connected by a line segment. Since U is connected, any two points can be joined by a finite sequence of such segments. Thus, $\tilde{u}$ is constant everywhere in U. As $u = \tilde{u}$ almost everywhere, u is constant almost everywhere in U.

Case 2: $1 \le p \le n$ (The General Case using Mollifiers)

When $p \le n$, the Morrey inequality doesn't apply, so we don't automatically get a continuous representative. We must use a different technique.

1. Let η_ϵ be the standard mollifier. For any compactly contained open subset $V \Subset U$, we can define the mollification of u, denoted $u_\epsilon = u * \eta_\epsilon$, for ϵ small enough.

2. A key property of mollification is that it commutes with differentiation:

$$Du_\epsilon = D(u * \eta_\epsilon) = (Du) * \eta_\epsilon$$

3. Since we are given that $Du = 0$ a.e. in U, its mollification is also zero:

$$(Du) * \eta_\epsilon(x) = \int_U Du(y)\eta_\epsilon(x - y)\, dy = \int_U 0 \cdot \eta_\epsilon(x - y)\, dy = 0$$

This holds for all $x \in V$.

4. The function u_ϵ is smooth (C^∞). Since its gradient Du_ϵ is zero everywhere in V, and V is connected, classical calculus tells us that u_ϵ must be a constant on V. Let's call this constant c_ϵ.

5. We also know that as $\epsilon \to 0$, $u_\epsilon \to u$ in $L^p(V)$. The sequence of constant functions $\{u_\epsilon\}$ converges in $L^p(V)$ to u. A sequence of constant functions can only converge in L^p if the constants themselves converge. Let $c = \lim_{\epsilon\to 0} c_\epsilon$. The limit function must be the constant function c.

6. Therefore, $u = c$ almost everywhere in V. Since this holds for any arbitrary compactly contained connected open set $V \subset U$, and U itself is connected, the constant c must be the same for all such sets. Thus, u is constant almost everywhere in U.

---

14. Verify that if $n > 1$, the unbounded function $u = \log\log\left(1 + \frac{1}{|x|}\right)$ belongs to $W^{1,n}(U)$, for $U = B^0(0,1)$.

From Solvely.ai

This question directly relates to our previous discussions. The initial problem contrasted the Sobolev and Morrey inequalities, which describe the properties of functions in Sobolev spaces like $W^{1,p}$. This problem provides a concrete example for the critical case where the integrability exponent equals the dimension ($p = n$), a case that lies on the boundary between the Sobolev and Morrey embedding theorems. It demonstrates a key feature of the space $W^{1,n}$: functions within it are not necessarily bounded, as this specific function $u(x)$ is unbounded near the origin.

To verify that $u(x) = \log\log(1 + \frac{1}{|x|})$ belongs to $W^{1,n}(U)$ for $U = B^0(0,1)$ and $n > 1$, we must show two things:

(a) $u \in L^n(U)$

(b) Its weak derivative, Du, exists and belongs to $L^n(U)$.

We will use polar coordinates, where $r = |x|$, and the volume element is $dx = r^{n-1}drd\sigma$, with $d\sigma$ being the surface element of the unit sphere S^{n-1}.

Step 1: Show $u \in L^n(U)$

We need to prove that $\int_U |u(x)|^n dx < \infty$.

$$\int_U |u(x)|^n dx = \int_{S^{n-1}} \int_0^1 |\log\log(1 + \frac{1}{r})|^n r^{n-1} dr d\sigma$$

Let ω_n be the surface area of S^{n-1}. The integral becomes:

$$\omega_n \int_0^1 (\log\log(1 + \frac{1}{r}))^n r^{n-1} dr$$

To analyze the convergence, we examine the integrand's behavior as $r \to 0$. For small r, $1/r$ is large, and $\log(1 + 1/r) \approx \log(1/r)$. Therefore, the function u behaves like $\log(\log(1/r))$. The integrand behaves like $(\log(\log(1/r)))^n r^{n-1}$. The polynomial term r^{n-1} (since $n > 1$) goes to zero much faster than the double-logarithmic term $(\log(\log(1/r)))^n$ grows. Specifically, for any

$$\alpha > 0, \lim_{r \to 0^+} r^\alpha \left(\log(\log(1/r))\right)^n = 0$$

This means the integrand is bounded on the interval $[0,1]$, and thus the integral is finite. So, $u \in L^n(U)$.

Step 2: Show $Du \in L^n(U)$

For $x \neq 0$, u is smooth, so its weak derivative coincides with its classical derivative. Let $f(r) = \log\log(1 + 1/r)$. Using the chain rule, $Du = f'(r)\frac{x}{r}$.

$$f'(r) = \frac{d}{dr}\log(\log(1 + 1/r)) = \frac{1}{\log(1 + 1/r)} \cdot \frac{1}{1 + 1/r} \cdot \left(-\frac{1}{r^2}\right) = -\frac{1}{r(r+1)\log(1 + 1/r)}$$

So, $|Du(x)| = |f'(r)| = \frac{1}{r(r+1)\log(1+1/r)}$.

Now, we must check if $\int_U |Du(x)|^n dx$ is finite.

$$\begin{aligned}\int_U |Du(x)|^n dx &= \omega_n \int_0^1 \left(\frac{1}{r(r+1)\log(1+1/r)}\right)^n r^{n-1} dr \\ &= \omega_n \int_0^1 \frac{1}{r^n(r+1)^n(\log(1+1/r))^n} r^{n-1} dr \\ &= \omega_n \int_0^1 \frac{1}{r(r+1)^n(\log(1+1/r))^n} dr\end{aligned}$$

Let's analyze the integrand near the singularity at $r = 0$. As $r \to 0$, $(r+1)^n \to 1$ and $\log(1+1/r) \approx \log(1/r)$. The integrand behaves like:

$$\frac{1}{r(\log(1/r))^n}$$

To check for convergence, we can use a substitution. Let $s = \log(1/r)$, so $r = e^{-s}$ and $dr = -e^{-s}ds$. As $r \to 0^+$, $s \to \infty$.

$$\int \frac{1}{r(\log(1/r))^n} dr = \int \frac{1}{e^{-s}s^n}(-e^{-s}ds) = -\int \frac{1}{s^n} ds = -\int s^{-n} ds$$

Since we are given $n > 1$, the integral is:

$$= -\frac{s^{-n+1}}{-n+1} + C = \frac{1}{(n-1)s^{n-1}} + C = \frac{1}{(n-1)(\log(1/r))^{n-1}} + C$$

Evaluating the definite integral from a small $a > 0$ to $\delta < 1$:

$$\lim_{a\to 0^+} \left[\frac{1}{(n-1)(\log(1/r))^{n-1}}\right]_a^\delta = \frac{1}{(n-1)(\log(1/\delta))^{n-1}} - \lim_{a\to 0^+} \frac{1}{(n-1)(\log(1/a))^{n-1}}$$

As $a \to 0^+$, $\log(1/a) \to \infty$, so the limit term is 0. The integral is therefore convergent. Since both $\int_U |u|^n dx$ and $\int_U |Du|^n dx$ are finite, we have successfully verified that $u \in W^{1,n}(U)$.

AI Questions and Responses

How are Hölder Spaces and Lebesque Spaces related to each other?

To understand the relationship between Hölder spaces and Lebesgue spaces, it is necessary to examine their definitions, the domains on which they are defined, and the embedding theorems that connect them.

Step-by-Step Explanation

1. **Definitions of the Spaces**
   - **Lebesgue Spaces (L^p):** These spaces measure the integrability of a function. For a measurable set $\Omega \subset \mathbb{R}^n$, a function f belongs to $L^p(\Omega)$ if the integral of its absolute value raised to the power p is finite: $\|f\|_{L^p} = (\int_\Omega |f|^p dx)^{1/p} < \infty$. These spaces focus on "size" and "averaging" behavior.

- **Hölder Spaces ($C^{k,\alpha}$):** These spaces measure the regularity or smoothness of a function. A function f is in the Hölder space $C^{0,\alpha}(\Omega)$ (often denoted C^α) if it is continuous and satisfies a specific growth condition: $|f(x) - f(y)| \leq C|x - y|^\alpha$ for all $x, y \in \Omega$, where $0 < \alpha \leq 1$. These spaces focus on"pointwise" behavior and continuity.

2. **Inclusion on Bounded Domains**

   On a bounded domain Ω with a sufficiently smooth boundary, there is a hierarchical relationship based on the "strength" of the norm:

   - **Hölder implies Lebesgue:** Any function that is Hölder continuous on a bounded domain is also bounded (since Ω is compact). Because bounded functions on bounded domains are integrable to any power p, we have the embedding $C^{0,\alpha}(\Omega) \subset L^p(\Omega)$ for all $1 \leq p \leq \infty$.
   - **Lebesgue does not imply Hölder:** A function in L^p can be discontinuous or even unbounded at certain points (e.g., $f(x) = x^{-1/4}$ is in $L^2(0, 1)$ but is not continuous, let alone Hölder continuous).

3. **Sobolev Embedding Theorems (The Bridge)** The primary mathematical link between these two spaces is provided by ***Sobolev Spaces*** ($W^{k,p}$). Sobolev spaces act as an intermediary by requiring a function and its derivatives to be in L^p.

   - **Morrey's Inequality:** If a function f is in the Sobolev space $W^{1,p}(\Omega)$ and $p > n$ (where n is the dimension of the space), then f is actually Hölder continuous. Specifically, $W^{1,p}(\Omega) \hookrightarrow C^{0,\gamma}(\Omega)$ where $\gamma = 1 - n/p$.
   - This means that if a function has "enough" integrability of its derivatives (Lebesgue property), it gains a certain level of smoothness (Hölder property).

4. **Comparison of Properties**

Feature	Lebesgue Spaces (L^p)	Hölder Spaces (C^α)
Focus	Global integrability / "Size"	Local smoothness / Regularity
Elements	Equivalence classes of functions	Actual continuous functions
Sensitivity	Insensitive to values at single points	Highly sensitive to pwise variations
Relationship	L^p is a "larger" space	C^α is a more restrictive space

Final Answer Summary

The relationship between Hölder spaces (C^α) and Lebesgue spaces (L^p) is characterized by the following:

1. ****Inclusion:**** On bounded domains, Hölder spaces are subsets of Lebesgue spaces ($C^\alpha \subset L^p$). A function with Hölder regularity is necessarily integrable, but an integrable function is not necessarily Hölder continuous.
2. ****Regularity vs. Integrability:**** Lebesgue spaces categorize functions by their "bulk" size and integrability, while Hölder spaces categorize them by their "smoothness" and pointwise behavior.
3. ****Sobolev Connection:**** They are linked via Sobolev Embedding Theorems. If a function's derivatives are in a Lebesgue space L^p with p sufficiently large ($p > n$), the function is guaranteed to be in a Hölder space.
4. ****The L^∞ Limit:**** As $p \to \infty$, the Lebesgue space L^∞ consists of essentially bounded functions. While C^α functions are in L^∞, L^∞ functions are not necessarily in C^α unless they possess specific continuity properties.

Final Answer:

1. Hölder spaces $C^\alpha(\Omega)$ are subsets of Lebesgue spaces $L^p(\Omega)$ on bounded domains: every Hölder continuous function is also Lebesgue integrable, but not every Lebesgue integrable function is Hölder continuous; they are connected via Sobolev embedding theorems, which state

that sufficient integrability of derivatives (in L^p) can imply Hölder continuity.¡/strong¿ Relationship between Hölder and Lebesgue spaces Defining Lebesgue and Hölder spaces Lebesgue spaces $L^p(\Omega)$ consist of functions whose absolute value raised to the p-th power is integrable over Ω. Hölder spaces $C^\alpha(\Omega)$ contain functions that satisfy a specific continuity condition: for all x, y, $|f(x)-f(y)| \leq C|x-y|^\alpha$ with $0 < \alpha \leq 1$. Thus, Hölder spaces focus on regularity, while Lebesgue spaces focus on integrability. Inclusion and hierarchy on bounded domains Every Hölder continuous function on a bounded domain is bounded, and therefore integrable for any p. This gives the inclusion $C^\alpha(\Omega) \subset L^p(\Omega)$ for all $1 \leq p \leq \infty$. However, not every function in $L^p(\Omega)$ is Hölder continuous—integrability does not imply smoothness. Connecting spaces via Sobolev embeddings Sobolev spaces $W^{k,p}(\Omega)$ require derivatives of a function to be in L^p. When $p > n$ (dimension), Sobolev embedding theorems (e.g., Morrey's inequality) guarantee that these functions are Hölder continuous: $W^{1,p}(\Omega) \hookrightarrow C^{0,\gamma}(\Omega)$, where $\gamma = 1 - n/p$. Thus, sufficient integrability of derivatives yields Hölder regularity. Distinguishing properties of the spaces Lebesgue spaces classify functions based on their global size, while Hölder spaces classify based on local continuity. Lebesgue functions can be highly irregular, but Hölder functions are smooth. Hölder spaces are more restrictive subsets within Lebesgue spaces. Answer: ¡strong¿Hölder spaces $C^\alpha(\Omega)$ are subsets of Lebesgue spaces $L^p(\Omega)$ on bounded domains: every Hölder continuous function is also Lebesgue integrable, but not every Lebesgue integrable function is Hölder continuous; they are connected via Sobolev embedding theorems, which state that sufficient integrability of derivatives (in L^p) can imply Hölder continuity.

Extension Theorem

How helpful is the Extension Theorem to PDE?

The Extension Theorem is a fundamental tool in the theory of Partial Differential Equations (PDEs), particularly when dealing with Sobolev spaces and boundary value problems. Its primary utility lies in allowing mathematicians to treat functions defined on a restricted domain as if they were defined on the entire Euclidean space.

****Step-by-Step Explanation of Utility****

1. **Overcoming Domain Restrictions** In PDE theory, many powerful analytical tools—such as the Fourier Transform, convolution, and certain fundamental solutions—are naturally defined on the entire space $\mathbb{R}^n$. However, physical problems are usually confined to a bounded domain Ω. The Extension Theorem allows a function $u \in W^{k,p}(\Omega)$ to be extended to a function $\bar{u} \in W^{k,p}(\mathbb{R}^n)$ such that $\bar{u} = u$ within Ω. This enables the application of global analytical techniques to local problems.
2. **Establishing Sobolev Embedding Theorems** One of the most critical uses of the extension theorem is in proving embedding theorems (like the Sobolev Inequality or Rellich-Kondrachov Compactness Theorem) for bounded domains.
   - ****The Logic:**** It is often easier to prove that $W^{k,p}(\mathbb{R}^n) \hookrightarrow L^q(\mathbb{R}^n)$ using global estimates.
   - ****The Application:**** By extending a function from Ω to $\mathbb{R}^n$, applying the global embedding, and then restricting the result back to Ω, we prove that the same embedding holds for the bounded domain, provided the boundary $\partial\Omega$ is sufficiently smooth (usually Lipschitz or C^k).
3. **Handling Boundary Regularity** The theorem is essential for defining and understanding "traces" (the values of a function on the boundary). To analyze how a solution behaves near

the boundary, one often extends the function across the boundary into a larger neighborhood. This simplifies the process of performing integration by parts and establishing energy estimates near $\partial\Omega$.

4. **Numerical Analysis and Approximation** In the context of Finite Element Methods (FEM) and other numerical schemes, extension theorems justify the approximation of functions. They ensure that functions in Sobolev spaces can be approximated by smooth functions (C^∞) up to the boundary, which is a prerequisite for many convergence proofs.
5. ****Summary of Importance****

Feature	Benefit in PDE Analysis
Global Tools	Permits the use of Fourier Transforms and convolutions on bounded domains.
Embeddings	Essential for proving $W^{k,p}(\Omega) \subset L^q(\Omega)$ and Hölder continuity.
Trace Theory	Provides the mathematical framework for non-homogeneous boundary conditions.
Regularity	Allows for the "straightening of the boundary" to study local solution behavior.

****Final Answer Summary**** The Extension Theorem is **highly helpful and indispensable** in PDE theory. It bridges the gap between functions defined on specific geometries and the powerful analytical machinery available for $\mathbb{R}^n$. Without it, proving the existence, uniqueness, and regularity of solutions in Sobolev spaces for bounded domains would be significantly more difficult, if not impossible, for many classes of equations.

Final Answer: 1. The extension theorem is highly helpful and indispensable in PDEs, as it enables the application of global analytical techniques (like Sobolev embeddings, Fourier analysis, and trace theory) to functions defined on bounded domains, thereby greatly simplifying the proofs of existence, regularity, and boundary behavior of solutions.¡/strong¿ Role and utility of the Extension Theorem in PDE theory Enables global analytical techniques on local domains The extension theorem allows a function in a Sobolev space on a bounded domain Ω to be extended to the whole space $\mathbb{R}^n$ while preserving its Sobolev regularity, thus making global tools like Fourier transforms and convolutions available for local PDE problems. Facilitates Sobolev embedding and compactness results By extending a function from Ω to $\mathbb{R}^n$, global Sobolev embedding and compactness theorems can be applied and then restricted back to Ω, allowing key regularity and continuity results to hold for domains with sufficient boundary regularity. Aids in boundary regularity and trace theorems The theorem permits extension of functions across the boundary, simplifying the study of traces (boundary values) and making integration by parts and energy estimates tractable near $\partial\Omega$. Supports numerical approximation and smooth approximations The ability to extend Sobolev functions ensures they can be approximated by smooth functions in numerical schemes, such as finite element methods, which is essential for convergence and error analysis. Answer: ¡strong¿The extension theorem is highly helpful and indispensable in PDEs, as it enables the application of global analytical techniques (like Sobolev embeddings, Fourier analysis, and trace theory) to functions defined on bounded domains, thereby greatly simplifying the proofs of existence, regularity, and boundary behavior of solutions.

Motivation for Hölder Norm?

What is the motivation for Hölder Norm?

Great question — the Hölder norm isn't just a technical definition; it's designed to *****quantify how smooth a function is*****, in a way that sits naturally **between mere continuity and differentiability**.

Let's build the motivation step by step.

—

1. The problem: continuity is too weak

   If a function is continuous, nearby points map to nearby values — but *how nearby* is uncontrolled.

   A function can be continuous and still oscillate wildly. For many areas (PDEs, geometry, numerical analysis), we need **quantitative control** on how fast a function changes.

   The Lipschitz condition improves this:

   $$|u(x) - u(y)| \leq C|x - y|.$$

   This says the function grows at most linearly with distance. But Lipschitz continuity is often **too strong** — many important solutions to PDEs fail to be Lipschitz.

   So we want something in between.

   —

2. Hölder continuity: controlled roughness

   Hölder continuity relaxes the Lipschitz condition:

   $$|u(x) - u(y)| \leq C|x - y|^\alpha, \quad 0 < \alpha < 1.$$

   Now the function is allowed to be rougher, but not arbitrarily rough.

   Interpretation:

   - smaller $(\alpha) \to$ rougher behavior allowed
   - $(\alpha = 1) \to$ Lipschitz continuity
   - $(\alpha > 0) \to$ stronger than plain continuity

   This captures **fractional smoothness** — a key idea in modern analysis.

   —

3. Why define the Hölder *norm*?

   We want to measure both:

   (a) Size of the function

   $$\sup |u|$$

   (b) Smoothness (oscillation control)

   $$[u]_{C^{0,\alpha}} = \sup_{x \neq y} \frac{|u(x) - u(y)|}{|x - y|^\alpha}.$$

   The Hölder norm combines them:

   $$|u|_{C^{0,\alpha}} = \sup |u| + [u]_{C^{0,\alpha}}.$$

   Motivation:

   - makes the space **complete** (Banach space)

- allows convergence + compactness arguments
- controls oscillation at *all scales*

This scale-sensitive control is exactly what many PDE estimates require.

4. Geometric intuition

   Think of zooming in on the graph of the function.

   - continuous → graph eventually settles down
   - Lipschitz → graph has bounded slope
   - Hölder → graph becomes flatter at a predictable rate

   The exponent (α) tells you **how fast oscillations shrink under magnification**.

   —

5. Why analysts love Hölder norms

   They naturally appear in:

   - PDE regularity theory
     Solutions to elliptic/parabolic equations often gain Hölder regularity even when derivatives fail to exist classically.
   - Compactness theorems
     Hölder bounds prevent wild oscillations → convergence control.
   - Interpolation
     They bridge spaces of different smoothness.
   - Fractional smoothness
     They anticipate Sobolev and Besov scale ideas.

   —

6. Big-picture motivation

   The Hölder norm is designed to:

   Quantify smoothness in a scale-sensitive way that is strong enough for analysis but flexible enough for real-world solutions.

   It measures ***how controlled a function's oscillation is***, not just whether it's continuous.

—

If you want, I can next explain:

- Why Hölder spaces appear naturally in elliptic PDE estimates
- Relationship to Lipschitz and Sobolev norms
- Geometric intuition with graphs
- Why interpolation inequalities use Hölder norms
- Examples of Hölder but non-Lipschitz functions
- Why Hölder regularity emerges from smoothing

Just say what direction you want.

Exercises for mollification

Provide 5 exercises to understand mollification

The request for exercises on mollification is directly related to our previous discussion. In the last step, we constructed a smooth cutoff function using a technique called mollification. This technique is fundamental in the theory of Sobolev spaces, which is the natural setting for the Sobolev and Morrey inequalities we discussed initially. Understanding mollification is key to proving many results in this area, such as the density of smooth functions in Sobolev spaces and the properties of weak solutions to PDEs.

Here are 5 exercises to help you better understand how to use mollification:

Exercise 1: Approximation of Identity

Let η be a standard mollifier and define $\eta_\epsilon(x) = \frac{1}{\epsilon^n}\eta(\frac{x}{\epsilon})$ for $\epsilon > 0$. Let $f \in L^p(\mathbb{R}^n)$ for $1 \leq p < \infty$. Show that the convolution $f_\epsilon = f * \eta_\epsilon$ converges to f in the L^p norm as $\epsilon \to 0$. That is, prove:

$$\lim_{\epsilon \to 0} \|f * \eta_\epsilon - f\|_{L^p} = 0$$

Hint: Use Minkowski's integral inequality and the fact that translations are continuous in L^p.

—

$$\begin{aligned}
\|u\|_p &= \left(\sum_i u_i^p\right)^{1/p} \\
\|f * \eta_\epsilon - f\|_{L^p} &= \left(\sum_i (\,(f * \eta_\epsilon)(x_i) - f(x_i)\,)^p\right)^{1/p} \\
&\leq \left(\sum_i (f * \eta_\epsilon)^p\right)^{1/p} - \left(\sum_i f^p\right)^{1/p} \\
&\leq \|f * \eta_\epsilon\|_p - \|f\|_p \\
\lim_{\epsilon \to 0} \|f * \eta_\epsilon - f\|_{L^p} &\leq \lim_{\epsilon \to 0} \|f * \eta_\epsilon\|_p - \lim_{\epsilon \to 0} \|f\|_p \\
&\leq \|f\|_p - \|f\|_p \\
&\leq 0
\end{aligned}$$

The norm must be positive therefore the limit is zero.

Exercise 2: Density of Smooth Functions in L^p

Prove that the space of compactly supported smooth functions, $C_c^\infty(\mathbb{R}^n)$, is dense in $L^p(\mathbb{R}^n)$ for $1 \leq p < \infty$.

Hint: Use the result from Exercise 1. First, show that for any $f \in L^p$, its mollification $f_\epsilon = f * \eta_\epsilon$ is a smooth function. Then, consider approximating f by functions with compact support, for example $f_k = f \cdot \chi_{B(0,k)}$, where χ is the characteristic function for the ball of radius k. Combine these two approximation steps.

—

Exercise 3: Mollification and Derivatives

Let $u \in W^{1,p}(U)$, where U is an open set in $\mathbb{R}^n$. Let $u_\epsilon = u * \eta_\epsilon$ be the mollification of u (assuming u is extended by zero outside U). Show that the derivative of the mollified function is the mollification of the weak derivative. Specifically, for any multi-index α with $|\alpha| = 1$, prove that:

$$D^\alpha(u * \eta_\epsilon) = (D^\alpha u) * \eta_\epsilon$$

Hint: Use the definition of weak derivative and properties of convolution.

—

Exercise 4: A Product Rule for Sobolev Functions

Let $u, v \in W^{1,p}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$ for $1 \leq p < \infty$. Prove that their product uv is in $W^{1,p}(\mathbb{R}^n)$ and that the Leibniz rule (product rule) holds for their weak derivatives:

$$D(uv) = (Du)v + u(Dv)$$

Hint: First, assume u, v are smooth. Then, for general u, v, consider their mollifications u_ϵ and v_ϵ. Show that $u_\epsilon v_\epsilon \to uv$ in L^p and $D(u_\epsilon v_\epsilon)$ converges in L^p to $(Du)v + u(Dv)$.

—

Exercise 5: Friedrichs' Inequality

Let U be a bounded open set in $\mathbb{R}^n$. Prove that there exists a constant $C > 0$ such that for any function $u \in C_c^1(U)$ (smooth functions with compact support in U), the following inequality holds:

$$\|u\|_{L^2(U)} \leq C\|\nabla u\|_{L^2(U)}$$

This shows that for functions that are zero on the boundary, the L^2 norm of the gradient controls the L^2 norm of the function itself. This is a foundational result for the Poincaré inequality.

Hint: For $u \in C_c^1(U)$, write $u(x) = \int_{-\infty}^{x_i} \frac{\partial u}{\partial x_i}(x_1, ..., t, ..., x_n)dt$. Apply the Cauchy-Schwarz inequality to this integral representation.

Exercises for Convolusions.

Provide 5 exercises that help to understand Convolusions.

This request for exercises on convolutions is directly related to our previous discussions. We used convolution with a mollifier to construct a smooth cutoff function, and the exercises on mollification heavily relied on the properties of convolution. Convolutions are a fundamental tool in the analysis of Sobolev spaces and PDEs, which is the context for the Hölder, Sobolev, and Morrey inequalities we started with.

Here are 5 exercises to help you better understand how to use convolutions:

Exercise 1: Basic Properties of Convolution

Let f, g, h be functions in $L^1(\mathbb{R}^n)$. Prove the following properties of the convolution operation $*$:

1. **Commutativity:** $f * g = g * f$
2. **Associativity:** $(f * g) * h = f * (g * h)$
3. **Distributivity:** $f * (g + h) = f * g + f * h$

Hint: These can be proven by direct computation using the definition of convolution, $(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x - y)dy$, and applying Fubini's theorem for associativity.

Commutatitivty use a change of variable

$$\begin{aligned}(f * g)(x) &= \int_{\mathbb{R}^n} f(y)g(x-y)\,dy \\ &= \int_{\mathbb{R}^n} f(x-z)g(z)\,dz && \text{where } z = y - x \\ &= (g * f)(x)\end{aligned}$$

Associativity

$$\begin{aligned}(h * (f * g))(x) &= \left(h * \int_{\mathbb{R}^n} f(y)g(x-y)\,dy\right) \\ &= \int_{\mathbb{R}^n} h(z) \int_{\mathbb{R}^n} f(y)g(x-y)\,dy\,dz \\ &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} h(z)f(y)g(x-z-y)\,dy\,dz \\ &= \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} h(w-y)f(y)g(x-w)\,dy\,dw && \text{where } w = z - y \\ &= \int_{\mathbb{R}^n} (h * f)(w)g(x-w)\,dy\,dw \\ &= ((h * f) * g)(x)\end{aligned}$$

Distributive Straight forward.

—

Exercise 2: Young's Inequality for Convolutions

Let $1 \le p, q, r \le \infty$ be such that $\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}$. If $f \in L^p(\mathbb{R}^n)$ and $g \in L^q(\mathbb{R}^n)$, prove that their convolution $f * g$ is in $L^r(\mathbb{R}^n)$ and satisfies the inequality:

$$\|f * g\|_{L^r} \le \|f\|_{L^p}\|g\|_{L^q}$$

Hint: This is a generalization of the simple case where $r = 1$ (which requires $p = 1, q = 1$). The proof involves a clever application of the Hölder inequality. This inequality is crucial for understanding how convolution interacts with L^p spaces.

Recall that the norm for $L^p(\mathbb{R}^n)$ is

$$\|u\|_{L^p} = \left(\int_{\mathbb{R}^n} u(x)^p \, dx \right)^{1/p}$$

and Hölder's Inequality

$$\int |fg| \leq \|f\|_{L^p} \|g\|_{L^q} \text{ where } \frac{1}{p} + \frac{1}{q} = 1$$

Direct application gives

$$\begin{aligned}
\|f * g\|_{L^r} &\leq \|f\|_{L^p} \|g\|_{L^q} \\
\left\| \int f(y) g(x-y) dy \right\|_{L^r} &\leq \left(\int_{\mathbb{R}^n} f(x)^p \, dx \right)^{1/p} \left(\int_{\mathbb{R}^n} g(x)^p \, dx \right)^{1/q}
\end{aligned}$$

$$\begin{aligned}
\left\| \int f(y) g(x-y) dy \right\|_{L^r} &= \left(\int \left| \int f(y) g(x-y) \, dy \right| \, dx \right)^{1/r} \\
&\leq \int \left(\left| \int f(y) g(x-y) \, dy \right| \right)^{1/r} dx \\
&\leq \left(\int f(x) \, dx \right)^{1/r} \left(\int g(x) \, dx \right)^{1/r}
\end{aligned}$$

—

Exercise 3: Convolution and the Fourier Transform

Let $f, g \in L^1(\mathbb{R}^n)$. The Fourier transform of a function h is defined as $\hat{h}(\xi) = \int_{\mathbb{R}^n} h(x) e^{-2\pi i x \cdot \xi} dx$. Prove that the Fourier transform of a convolution is the pointwise product of the Fourier transforms:

$$\widehat{f * g}(\xi) = \hat{f}(\xi) \hat{g}(\xi)$$

Hint: Write out the definition of the Fourier transform for $f * g$ and use Fubini's theorem to switch the order of integration. This property is one of the most important aspects of the Fourier transform and is why convolution is used to model filtering processes.

—

Exercise 4: Solving a Differential Equation with Convolution

Consider the ordinary differential equation (ODE) $u'' - u = f$ on $\mathbb{R}$, where f is a continuous function with compact support. A solution can be found using the Green's function (or fundamental solution) $G(x) = -\frac{1}{2} e^{-|x|}$. Show that the function $u(x) = (G * f)(x)$ is a solution to the ODE.

Hint: You need to compute the first and second derivatives of the convolution $u(x) = -\frac{1}{2} \int_{-\infty}^{\infty} e^{-|x-y|} f(y) dy$. Be careful when differentiating the absolute value function. This exercise demonstrates how convolution provides a general method for solving linear, constant-coefficient differential equations.

—

Exercise 5: Support of a Convolution

Let f and g be functions on $\mathbb{R}^n$ with compact support. The support of a function h, denoted $\text{supp}(h)$, is the closure of the set where $h(x) \neq 0$. Prove that the support of the convolution $f * g$ is contained in the vector sum of their individual supports:

$$\text{supp}(f * g) \subseteq \text{supp}(f) + \text{supp}(g)$$

where $A + B = \{a + b \mid a \in A, b \in B\}$. This property is essential for localization arguments, such as the one we used to construct the smooth cutoff function in the previous answer.

Hint: Consider a point x where $(f * g)(x) \neq 0$. This means the integrand $f(y)g(x - y)$ must be non-zero for some y. What does this imply about y and $x - y$?

Describe the $H^n(H)$ Space.

Describe the $H^n(U)$ Space.

The notation $H^k(U)$ is central to our entire conversation. It represents a specific type of ***Sobolev space***, which is a fundamental concept in the study of partial differential equations and the inequalities we've been discussing.

Definition

For an integer $k \geq 0$, $H^k(U)$ is the Sobolev space $W^{k,2}(U)$. It is the set of all functions u defined on a domain U that satisfy two conditions:

1. The function u is in $L^2(U)$, meaning its square is integrable: $\int_U |u|^2 dx < \infty$.
2. All of its weak derivatives up to order k are also in $L^2(U)$.

More formally, using multi-index notation for derivatives:

$$H^k(U) = \{u \in L^2(U) \mid D^\alpha u \in L^2(U) \text{ for all multi-indices } |\alpha| \leq k\}$$

Here, $D^\alpha u$ denotes the weak derivative of u.

Why it's important to our conversation:

- **Special Case of $W^{k,p}$:**
  The inequalities we first discussed, like Sobolev and Morrey, apply to the general Sobolev spaces $W^{k,p}(U)$. The $H^k(U)$ spaces are the special but extremely important case where $p = 2$. This is the case that corresponds to ***Hilbert spaces***, which have an inner product and are often much easier to work with than general Banach spaces.
- **Hilbert Space Structure**:
  The space $H^k(U)$ is a Hilbert space equipped with an inner product, for example:

  $$(u, v)_{H^k} = \sum_{|\alpha| \leq k} \int_U (D^\alpha u)(D^\alpha v)\, dx$$

  The norm $\|u\|_{H^k} = \sqrt{(u, u)_{H^k}}$ is induced by this inner product. This structure is what allows us to use tools like the ***Cauchy-Schwarz inequality*** so effectively, as we did in the proofs for the interpolation and Hardy inequalities.
- **Examples from our discussion:**
  - $H^1(U)$**:**
    This is the space of functions whose value and first derivative are square-integrable. We used this space extensively when proving the integration by parts formula for Sobolev functions.
  - $H_0^1(U)$**:**
    This is a subspace of $H^1(U)$ where the functions have a trace of zero on the boundary. We used this condition to prove the interpolation inequality, as it made the boundary term in the integration by parts formula disappear.
  - $H^2(U)$**:**
    This is the space of functions whose value, first derivative, and second derivative are all square-integrable. This higher-order ***regularity*** was a prerequisite for the interpolation inequality $\|Du\|_{L^2}^2 \leq C\|u\|_{L^2}\|D^2u\|_{L^2}$, as the term $\|D^2u\|_{L^2}$ needed to be finite.

Define elliptical, hyperbolic and parabolic PDE

.

Explain the terms elliptical, hyperbolic and parabolic in the context of PDE

The classification into ***elliptic, parabolic, and hyperbolic PDEs*** is really about the *geometry and behavior of solutions*, not just algebra. The cleanest way to see it is through the ***second-order linear PDE***.

—

1. **The general second-order PDE**
   In two variables, consider:

$$A(x,y), u_{xx} + 2B(x,y), u_{xy} + C(x,y), u_{yy} + \text{lower-order terms} = 0$$

   The classification depends on the discriminant:

$$\boxed{B^2 - AC}$$

   —

2. **The three types**
   (a) **Elliptic:** $B^2 - AC < 0$
   Example:

$$u_{xx} + u_{yy} = 0 \quad \text{(Laplace equation)}$$

   Meaning:
   - No real characteristic directions
   - Behavior is ***spatially smooth and global***
   - Disturbances spread instantly everywhere

   Key properties:
   - Solutions are very smooth (analytic if coefficients are nice)
   - No propagation of waves
   - Boundary conditions determine everything

   Physical intuition:
   - Steady-state equilibrium
   - Example: temperature in a solid at equilibrium

   —

   (b) **Hyperbolic:** $B^2 - AC > 0$
   Example:

$$u_{tt} - c^2 u_{xx} = 0 \quad \text{(wave equation)}$$

   Meaning:
   - Two real characteristic directions
   - Information propagates along ***characteristic curves***

   Key properties:
   - Finite speed of propagation

- Signals travel like waves
- Sharp fronts and discontinuities can occur

Physical intuition:

- Vibrations, waves, signals
- Example: string vibration, sound waves

—

(c) Parabolic: $B^2 - AC = 0$

Example:

$$u_t - ku_{xx} = 0 \quad \text{(heat equation)}$$

Meaning:

- One repeated characteristic direction
- Intermediate between elliptic and hyperbolic

Key properties:

- Diffusion/smoothing over time
- Infinite propagation speed (but decays rapidly)
- Irreversible behavior (time direction matters)

Physical intuition:

- Heat flow, diffusion
- Information spreads but *dissipates*

—

3. Geometric interpretation (very important)

The classification comes from the quadratic form:

$$A, dx^2 + 2B, dx, dy + C, dy^2$$

This behaves like conic sections:

Type	Geometry	Analogy
Elliptic	Ellipse	No real directions
Hyperbolic	Hyperbola	Two directions
Parabolic	Parabola	One direction

So:

- ***Hyperbolic*** → real characteristic curves (wave propagation)
- ***Elliptic*** → no real characteristics (global coupling)
- ***Parabolic*** → degenerate case

—

4. Characteristics (the real heart of it)

Characteristic curves satisfy:

$$A(dy)^2 - 2B\,dx\,dy + C(dx)^2 = 0$$

- Hyperbolic → two real solutions → two families of curves
- Parabolic → one repeated solution
- Elliptic → no real solutions

These curves are where:

- information travels (hyperbolic)
- PDE reduces to ODEs

—

5. **Big-picture intuition**

   Elliptic:

   "Everything depends on everything else at once."

   - Global constraint
   - No notion of time evolution

   —

   Hyperbolic:

   "Information travels along paths."

   - Signals move with finite speed
   - Domain of dependence is finite

   —

   Parabolic:

   "Information diffuses and forgets its past."

   - Smooths out irregularities
   - Has a preferred time direction

—

6. **In higher dimensions**

   For:

$$\sum_{i,j} a_{ij} u_{x_i x_j}$$

   Look at eigenvalues of matrix $A = (a_{ij})$:

   - ***Elliptic***: all eigenvalues same sign
   - ***Hyperbolic***: mixed signs
   - ***Parabolic***: at least one zero eigenvalue

   —

7. **Why this matters**

   This classification determines:

   - What kind of boundary/initial conditions are appropriate
   - Whether solutions are smooth
   - How information propagates
   - Which methods work (Fourier, energy, maximum principle, etc.)

   —

If you want, I can connect this directly to:

- characteristics and change of variables,
- energy estimates,
- or how this classification appears on manifolds (since you've been working in that direction).

How do you derive a Green's Function?

How do you derive a Green's Function?

Deriving a Green's function is less about a single recipe and more about a *strategy tailored to the operator and domain.* That said, there is a clear general framework you can follow.

—

1. **What a Green's function is**

   Suppose you have a linear PDE operator L and want to solve

   $$Lu(x) = f(x), \quad x \in \Omega$$

   with some boundary conditions.
   The Green's function $G(x, \xi)$ is defined by

   $$L_x G(x, \xi) = \delta(x - \xi)$$

   (with the same boundary conditions in x).
   Once you have G, the solution is

   $$u(x) = \int_\Omega G(x, \xi), f(\xi), d\xi.$$

   So the whole problem reduces to: ***solve the PDE with a point source***.

   —

2. **General derivation strategy**

   Step 1: Identify the operator and domain

   - Example: $L = -\Delta$, or $L = \partial_t - \Delta$
   - Domain: $\mathbb{R}^n$, bounded domain, interval, etc.
   - Boundary conditions matter *a lot*

   —

   Step 2: Solve the fundamental solution (free space)

   First ignore boundaries and solve:

   $$LG(x, \xi) = \delta(x - \xi), \quad x \in \mathbb{R}^n$$

   This is often done using:

   - Fourier transform
   - Symmetry (radial solutions)
   - ODE reduction (in 1D)

   —

   Example: Laplacian in $\mathbb{R}^3$

   Solve:

   $$-\Delta G = \delta(x)$$

   Using symmetry, assume $G = G(r)$, $r = |x|$, then:

   $$\Delta G = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dG}{dr} \right)$$

   Solving gives:

   $$G(x) = \frac{1}{4\pi|x|}$$

   —

Step 3: Impose boundary conditions

If the domain is not all of $\mathbb{R}^n$, modify G so that:

- $G(x, \xi)$ satisfies boundary conditions in x

Methods:

- ***Method of images***
- ***Eigenfunction expansions***
- ***Separation of variables***

—

Example: Interval $[0, L]$

For:

$$-u'' = f(x), \quad u(0) = u(L) = 0$$

Green's function satisfies:

$$-G''(x, \xi) = \delta(x - \xi)$$

Solve piecewise:

- For $x < \xi$: $G = Ax$
- For $x > \xi$: $G = B(L - x)$

Apply:

- Continuity at $x = \xi$
- Jump condition:

$$G'(\xi^+) - G'(\xi^-) = -1$$

Result:

$$G(x, \xi) = \begin{cases} \frac{x(L-\xi)}{L}, & x < \xi \\ \frac{\xi(L-x)}{L}, & x > \xi s \end{cases}$$

—

Step 4: Verify the delta property

You must check:

- G is continuous
- Derivative has correct jump:

$$\text{jump in derivative} = 1$$

This ensures:

$$LG = \delta$$

—

Step 5: Build the solution

Once G is known:

$$u(x) = \int_\Omega G(x, \xi) f(\xi), d\xi$$

—

3. Key intuition

A Green's function is:

The response of the system to a point source.

So you can think:

- General forcing = superposition of point sources
- Solution = sum (integral) of responses

—

4. **Common methods summarized**

   1. ***Fourier transform (best for $\mathbb{R}^n$)***
      Turn PDE into algebra:

$$\hat{G}(k) = \frac{1}{L(k)}$$

   —

   2. ***Eigenfunction expansion***
      If $L\phi_n = \lambda_n \phi_n$, then:

$$G(x, \xi) = \sum_n \frac{\phi_n(x)\phi_n(\xi)}{\lambda_n}$$

   —

   3. ***Method of images***
      Used for simple domains (half-space, etc.)
      —

   4. ***Direct ODE approach (1D)***
      Solve piecewise and enforce jump conditions.
      —

5. **Subtle but important point**
   Green's functions are:

   - ***not unique*** unless boundary conditions are fixed
   - depend heavily on domain geometry

   —

6. **If you want to go deeper**
   Given your background (you've been working with tensors and PDEs), the next level is:

   - Green's functions on manifolds
   - Heat kernel as a Green's function
   - Connection with fundamental solutions and distributions

   —

If you want, I can:

- Derive one fully step-by-step (Laplace, heat, wave), or
- Show how Green's functions relate to weak solutions and distributions.

Explain Sobolev Embedding

Explain Sobolev Embedding

The ***Sobolev Embedding Theorem*** tells us that functions with weak derivatives in L^p automatically belong to better spaces—typically some L^q space, or even a Hölder continuous space.

It is one of the most important tools in PDE and functional analysis because it converts ***derivative information*** into ***regularity information***.

—

1. **The basic idea**

   Suppose $u \in W^{1,p}(U)$. This means:

   - $u \in L^p(U)$
   - its first weak derivatives are also in $L^p(U)$

   Then Sobolev embedding says that u often belongs to a larger space like:

   - $L^q(U)$ for some $q > p$,
   - or $C^{0,\alpha}(U)$,
   - or even $L^\infty(U)$,

   depending on the dimension n and the exponent p.

   —

2. **The standard $W^{1,p} \to L^q$ embedding**

   Let $U \subset \mathbb{R}^n$ be bounded and regular enough.
   If $1 \le p < n$, then

   $$W^{1,p}(U) \hookrightarrow L^q(U)$$

   for all

   $$1 \le q \le p^* := \frac{np}{n-p}.$$

   Here

   $$p^* = \frac{np}{n-p}$$

   is called the ***Sobolev conjugate exponent***.
   This means there exists $C > 0$ such that

   $$|u|_{L^q(U)} \le C|u|_{W^{1,p}(U)}.$$

   —

   Example: $H^1(U) = W^{1,2}(U)$
   If $n \ge 3$, then

   $$H^1(U) \hookrightarrow L^{2n/(n-2)}(U).$$

   That is,

   $$|u|_{L^{2n/(n-2)}} \le C|u|_{H^1}.$$

   This is exactly what was used in your PDE problem.

   —

3. Why the exponent $p^* = \frac{np}{n-p}$?

The exponent comes from balancing the integrability of u and its derivative.
Roughly:

- differentiating costs one power of length,
- integration in dimension n gives scaling by n,
- balancing these yields

$$\frac{1}{p^*} = \frac{1}{p} - \frac{1}{n}.$$

Solving:

$$p^* = \frac{np}{n-p}.$$

This is the largest L^q space you can embed into in general.

—

4. The three cases

—

Case 1: $p < n$

Then

$$W^{1,p}(U) \hookrightarrow L^q(U), \quad q \leq p^*.$$

This gives ***better integrability***.
Example:

$$H^1(U) \hookrightarrow L^6(U) \quad \text{if } n = 3.$$

—

Case 2: $p = n$

Then

$$W^{1,n}(U) \hookrightarrow L^q(U) \quad \text{for all } q < \infty,$$

but generally not into L^∞.
This is the borderline case.

—

Case 3: $p > n$

Then

$$W^{1,p}(U) \hookrightarrow C^{0,\alpha}(\overline{U}),$$

where

$$\alpha = 1 - \frac{n}{p}.$$

This means functions are Hölder continuous.

So if derivatives are sufficiently integrable, the function is actually continuous.

—

5. The special case $n = 2$

For $H^1(U)$ in dimension 2:

$$H^1(U) \hookrightarrow L^q(U) \quad \forall q < \infty.$$

This was exactly the fact used in part i of your boundary value problem.
Because 2 is the critical dimension for H^1:

$$\frac{2n}{n-2}$$

becomes infinite when $n = 2$.
So in 2D, H^1 embeds into every finite L^q.

—

6. Why this matters in PDE

Sobolev embedding lets us estimate nonlinear terms.
Example:

$$\int_U c(x)u(x)v(x), dx.$$

To control this with Hölder, we need $u, v \in L^q$.
Sobolev embedding gives:

$$u, v \in H_0^1(U) \implies u, v \in L^q(U),$$

which makes the bilinear form bounded.
This is why Lax–Milgram works.

—

7. Continuous embedding notation

The notation

$$W^{1,p}(U) \hookrightarrow L^q(U)$$

means:

1. Every $u \in W^{1,p}(U)$ lies in $L^q(U)$,
2. There is a constant C such that

$$|u|_{L^q} \leq C|u|_{W^{1,p}}.$$

So the inclusion map is continuous.

—

8. Compact embedding

A stronger result is the ***Rellich–Kondrachov theorem***:
If $q < p^*$, then

$$W^{1,p}(U) \hookrightarrow\hookrightarrow L^q(U),$$

meaning the embedding is ***compact***.
So bounded sequences in $W^{1,p}$ have convergent subsequences in L^q.
This is essential for nonlinear PDEs and variational methods.

—

9. Intuition

If the derivative of u is in L^p, then u cannot oscillate too wildly.

Greater derivative control gives greater regularity:

- weak derivative in $L^p \Rightarrow$ better integrability,
- stronger $p \Rightarrow$ continuity,
- even stronger $\Rightarrow$ differentiability.

Sobolev embedding quantifies that.

—

10. The two embeddings from your problem

- For $n = 2$:

$$H_0^1(U) \hookrightarrow L^q(U) \quad \forall q < \infty.$$

- For $n \geq 3$:

$$H_0^1(U) \hookrightarrow L^{2n/(n-2)}(U).$$

These are the precise embeddings needed to estimate

$$\int_U cuv.$$

—

If you'd like next, I can explain ***why*** $H_0^1(U) \hookrightarrow L^{2n/(n-2)}(U)$ holds by proving the Sobolev inequality in the H^1 case.